

Systematic Review

Bridging Two Worlds: Sensor Technologies and AI for Fruit Detection in Latin America and Beyond—A Scoping Review

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Abstract

This scoping review synthesizes 35 studies on sensor technologies and artificial intelligence for fruit detection, classification, and quality assessment, contrasting Latin American and international research traditions. The analysis suggests notable differences in technological approaches between the included Latin American and international studies: Latin American research tends to emphasize in developing accessible, practical solutions using classical computer vision and low-cost hardware, while international studies more frequently employ through deep learning architectures, multi-modal sensing, and complete robotic automation systems. Across the included studies, relatively limited attention was given to AI-assisted decision support for agricultural practitioners, insufficient consideration of inclusivity, and the scarce integration of environmental sustainability into intelligent sensing system design. The review identifies that only 8 of 35 studies originate from Latin America, suggesting an uneven geographical distribution of the available evidence. The included studies generally reported high accuracy values, yet these findings must be interpreted with caution given the reliance on curated datasets that may not represent real-world variability. The reviewed evidence suggests that future research may benefit not from one approach dominating the other, but from a thoughtful integration of complementary strategies, including knowledge transfer, edge computing democratization, and human-centered design. Overall, this review suggests that the ultimate goal extends beyond accuracy metrics to the transformation of agricultural practices that enhance food security, economic development, and environmental sustainability across the global agricultural landscape. It is important to note that this work does not propose or validate a new fruit detection algorithm but rather synthesizes and critically evaluates existing scientific evidence regarding sensor technologies and artificial intelligence applied to fruit detection and quality assessment.



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1. Introduction

The global agricultural industry stands at a technological crossroads, where the convergence of artificial intelligence, sensor technologies, and computer vision promises to transform traditional practices that have remained largely unchanged for generations [1]. Among the most promising applications of these technologies is the automated detection, classification, and quality assessment of fruits—a domain that holds immense economic significance for both developed and developing nations [2]. The ability to non-destructively evaluate fruit quality, detect defects, assess ripeness, and predict shelf life using sensor-based systems represents a paradigm shift from subjective, labor-intensive manual inspection to objective, rapid, and scalable automated solutions [3].

Computer vision, as a subfield of artificial intelligence, has emerged as a particularly powerful tool for fruit quality assessment. By enabling machines to interpret and analyze visual information from the world, computer vision systems can replicate—and, in many cases, surpass—human capabilities in detecting subtle defects, classifying fruits by size and color, and assessing ripeness with remarkable consistency [4]. When combined with advanced machine learning techniques, particularly deep learning architectures such as Convolutional Neural Networks (CNNs), You Only Look Once (YOLO), and Vision Transformers (ViTs), these systems achieve accuracy levels that approach or exceed 99% in many applications [5]. The integration of complementary sensing modalities—including near-infrared spectroscopy, hyperspectral imaging, and 3D scanning—has further expanded the capabilities of these systems, enabling assessment of internal quality attributes such as soluble solids content, internal defects, and chemical composition that were previously accessible only through destructive testing [6].

The importance of these technologies extends far beyond academic curiosity. Globally, the fruit industry faces mounting pressures to reduce post-harvest losses, ensure food safety, meet increasingly stringent quality standards, and optimize supply chain efficiency [7]. The Food and Agriculture Organization estimates that approximately one-third of all food produced for human consumption is lost or wasted annually, with fruits and vegetables experiencing particularly high loss rates due to their perishable nature [8]. Automated quality assessment systems offer a pathway to dramatically reduce these losses by enabling rapid, accurate sorting at multiple points along the supply chain—from harvest to processing to retail [9].

In Latin America, these technologies hold particularly transformative potential. The region is a major global exporter of fruits, with countries like Ecuador, Colombia, Mexico, Peru, and Chile supplying bananas, pineapples, passion fruit, avocados, grapes, and countless other fruits to markets worldwide [10]. The quality and consistency of these exports directly impact the economic livelihoods of millions of farmers, processors, and exporters [11]. However, the adoption of advanced sensor technologies and AI in Latin American agricultural sectors has been uneven, constrained by limited access to sophisticated equipment, computational resources, and technical expertise [12]. This technological gap, if left unaddressed, threatens to undermine the region's competitiveness in increasingly quality-conscious global markets [13].

Despite the rapid proliferation of research in this domain, the literature remains fragmented and poorly synthesized. Studies emerge from diverse disciplinary backgrounds—computer science, agricultural engineering, food technology, and robotics—each employing different methodologies, reporting different performance metrics, and addressing different crops and applications [14]. This fragmentation makes it difficult for researchers, practitioners, and policymakers to gain a comprehensive understanding of the state of the art, identify best practices, or recognize critical gaps in the evidence base. Furthermore, to the best of our knowledge, the stark contrast between research

conducted in technologically advanced nations and that produced in resource-constrained contexts—such as Latin America—has not been systematically examined, leaving important questions about technology transfer, contextual adaptation, and equitable access largely unexplored.

To address these gaps, this scoping review aims to synthesize the existing evidence on sensor technologies and artificial intelligence for fruit detection, classification, and quality assessment. By systematically mapping the literature and critically analyzing both Latin American and international contributions, we seek to explore the technological landscape, identify strengths and limitations of different approaches, and identify potential directions for a more integrated, accessible, and impactful research. This review moves beyond simple cataloging of techniques to critically examine how different contexts, resources, and priorities shape the development and deployment of fruit detection technologies. We analyze not only what has been reported but also areas that have received comparatively limited attention—particularly concerning inclusivity, sustainability, and the human dimensions of technology adoption.

To guide this scoping review, we propose the following overarching questions, which emerged from preliminary analysis of the literature and reflect the comparative, critical, and forward-looking orientation of our investigation.

- RQ1 (Technological Approaches): What sensor technologies and computational techniques are being employed for fruit detection, classification, and quality assessment across different geographical contexts, and what are their reported performance characteristics?
- RQ2 (Regional Contrasts): How do Latin American and international research traditions differ in their technological approaches, resource utilization, research priorities, and implementation contexts for fruit detection technologies?
- RQ3 (Agricultural Applications): Which fruit species and agricultural stages (pre-harvest vs. post-harvest) are addressed across the literature, and what patterns emerge regarding research priorities and economic contexts?
- RQ4 (Critical Gaps and Blind Spots): What potential research gaps emerge regarding AI-assisted decision support, inclusivity, environmental sustainability, human-centered system design, and the transition from laboratory prototypes to real agricultural deployment?

By addressing these questions, this scoping review aims to provide a comprehensive, forward-looking synthesis that informs both research and practice. We seek to move the conversation beyond the “what” of sensor technologies and AI to the “so what” and “for whom”—analyzing the technological, economic, and social dimensions of fruit detection technologies. Ultimately, we aim to understand not just whether these technologies can improve fruit quality assessment, but whether they may contribute to a more equitable, sustainable, and resilient agricultural future for both Latin America and the global community. Our review is guided by the premise that technology must serve human needs—enhancing the capabilities of agricultural professionals, supporting the livelihoods of farmers, and ensuring that the benefits of innovation are shared equitably across the global agricultural landscape.

In contrast to traditional systematic reviews that primarily synthesize evidence addressing narrowly defined research questions [15,16], this scoping review aims to provide a broader understanding of the technological ecosystem surrounding AI-based fruit detection. Readers should therefore interpret the present work as a comprehensive mapping of sensing technologies, computational approaches, regional research priorities, methodological gaps, and emerging opportunities rather than as a quantitative comparison of algorithmic performance. This broader perspective enables the identification of research trends and future directions that extend beyond conventional effectiveness-based evidence synthesis.

Finally, rather than developing a novel artificial intelligence model, this review critically analyzes published evidence describing existing sensing technologies, computational approaches, validation methodologies, and reported performance. Consequently, the conclusions presented herein should be interpreted as evidence synthesis rather than experimental validation of a newly proposed detection algorithm.

2. Related Work

The literature on sensor technologies and artificial intelligence for fruit detection and quality assessment has grown considerably, yielding numerous reviews that have advanced our understanding of specific subdomains such as deep learning architectures, computer vision systems, ripeness classification, and disease detection. Table 1 presents a summary of these prior works, outlining their main goal, principal findings, and—crucially—how they diverge from the scope of our study. This comparison makes evident that, although existing reviews contribute meaningful perspectives on overarching trends, technological frameworks, or particular systems, they generally focus on algorithmic performance and technological implementation while giving comparatively limited attention to regional disparities, human-centered AI, inclusivity, deployment readiness, explainable decision support, and sustainability, which constitute the central analytical dimensions of the present scoping review.

Table 1. Summary of recent literature reviews on sensor technologies and artificial intelligence for fruit detection, classification, and quality assessment, highlighting scope, focus, and differences with the present review.

Authors	Goal	Main Findings	Differences with Our Review
Chandra et al. [17]	To review and propose image processing techniques for fruit classification based on shape, color, and texture features using SVM and feature extraction methods (HSV, DWT, GLCM).	SVM classification using color, texture, and shape features achieved up to 100% accuracy for some fruits (e.g., using Hue component); color+texture fusion improved classification to 97.2% for pomegranates.	This is a short review paper with limited empirical validation; it does not cover deep learning, sensor fusion, robotics, or regional comparisons; our scoping review systematically synthesizes 35 studies with a focus on Latin America vs. international approaches, inclusivity, sustainability, and critical gaps.
Ukwuoma et al. [18]	To conduct a comprehensive review of deep learning techniques for fruit detection and classification, and provide experimental implementation of CNN and ResNet-50 on the Fruit 360 dataset to guide beginners.	Deep learning models (especially CNNs and transfer learning) achieve > 95% accuracy; Fruit 360 is the most used dataset but lacks background variability; transfer learning and data augmentation (e.g., GANs) are key solutions for limited datasets; adversarial attacks can significantly reduce model accuracy; their CNN from scratch achieved 95% accuracy, while ResNet-50 achieved 99%.	This is a general survey of deep learning for fruit classification/detection without regional focus; it does not compare Latin American vs. international research, does not address sensor technologies (e.g., hyperspectral, NIR, 3D), lacks analysis of inclusivity, sustainability, or human-centered design, and does not examine post-harvest vs. pre-harvest applications or agricultural export contexts as our scoping review does.
Rojas Santelices et al. [19]	To conduct a systematic literature review (PRISMA-based) of artificial vision systems for fruit inspection and classification, identifying application fields, hardware configurations, and image processing algorithms used between 2015–2024.	RGB cameras (84%) and LED lighting dominate; deep learning (especially CNNs like ResNet, VGG, MobileNet) has surpassed traditional methods since 2020; main application areas: orchards, processing lines, and retail; preprocessing (filtering, augmentation), segmentation (Otsu, Canny, Watershed), and feature extraction are critical; transfer learning and data augmentation are key for generalization; 74% of studies used proprietary datasets.	This is a general SLR on computer vision for fruit inspection without regional focus; it does not compare Latin American vs. international research, does not address inclusivity, sustainability, or human-centered design, lacks analysis of AI-assisted decision support, does not examine the research–practice gap or technology transfer disparities, and does not focus on post-harvest vs. pre-harvest economic contexts as our scoping review does.
Sawant et al. [20]	To provide a comprehensive review of machine learning and deep learning techniques for fruit variety classification, including motivation, challenges, frameworks, and a practical CNN implementation (ResNet50) for mango variety classification.	CNN-based models (especially transfer learning with ResNet50, InceptionV3) achieve > 99% accuracy on curated datasets; key challenges include limited dataset availability, variability in fruit appearance (lighting, orientation, ripeness), and lack of commercialization; data augmentation and transfer learning are critical for improving generalization; their ResNet50 implementation achieved 99.7% testing accuracy on 8 mango varieties.	This is a general review focused on variety classification without regional or comparative analysis; it does not address Latin America vs. international research, lacks analysis of sensor technologies (hyperspectral, NIR, 3D), does not examine inclusivity, sustainability, human-centered design, or AI-assisted decision support, and does not cover the research–practice gap or technology transfer disparities as our scoping review does.

Table 1. *Cont.*

Authors	Goal	Main Findings	Differences with Our Review
Dhiman et al. [21]	To systematically review machine learning, deep learning, and statistical techniques for the detection, prediction, and classification of citrus fruit diseases (78 papers, 1995–2020).	SVM is the best ML technique; CNN is the best DL technique; PCA is the most used statistical method; RGB color space and color features dominate; accuracy is the primary evaluation metric; hybrid methods (e.g., PCA-ANN, CNN-fuzzy) improve accuracy; surface defects and canker are the most studied diseases; hyperspectral imaging and thresholding are widely used for image acquisition and processing.	This is a disease-specific SLR focused exclusively on citrus fruits; it does not address fruit detection/classification for quality assessment or variety classification across multiple fruit types; lacks regional comparison (Latin America vs. international), does not examine inclusivity, sustainability, human-centered design, AI-assisted decision support, or the research–practice gap; does not cover post-harvest vs. pre-harvest economic contexts or sensor technology democratization as our scoping review does.
Rizzo et al. [22]	To survey fruit ripeness classification methods, covering feature descriptors (color, volatiles, spectroscopy, fluorescence, spectral imaging) and statistical, machine learning, and deep learning approaches (2014–2022).	Deep learning (especially CNNs with transfer learning) dominates and achieves state-of-the-art accuracy without manual feature engineering; color (CIELAB, RGB, HSV) is the most used feature; non-destructive techniques (VNIR spectroscopy, hyperspectral/multispectral imaging) are key for in-field assessment; DL models like ResNet, VGG, and Mask R-CNN achieve > 95% accuracy; public datasets (Fruit 360, DeepFruits, etc.) are limited; interpretability and real-time deployment remain challenges.	This is a ripeness-specific survey without regional comparison; it does not address Latin America vs. international research, lacks analysis of inclusivity, sustainability, human-centered design, or AI-assisted decision support; does not cover post-harvest vs. pre-harvest economic contexts, sensor technology democratization, or the research–practice gap as our scoping review does.
Chaudhari and Waghmare [23]	To provide a review of machine vision-based fruit classification and grading techniques, covering preprocessing, segmentation, feature extraction (color, size, shape, texture using SURF, HOG, LBP), and ML/DL classifiers (KNN, SVM, ANN, CNN).	Color is the most important feature; HIS color space is widely used; SVM and LS-SVM achieve high accuracy; CNN and deep learning outperform traditional ML methods; manual inspection is time-consuming and error-prone; automated systems can reduce labor and improve efficiency; challenges include fruit variability and dataset limitations.	This is a general, short review paper without systematic methodology or regional focus; it does not compare Latin America vs. international research, lacks analysis of sensor technologies (hyperspectral, NIR, 3D), does not address inclusivity, sustainability, human-centered design, AI-assisted decision support, or the research–practice gap; does not cover post-harvest vs. pre-harvest economic contexts or technology democratization as our scoping review does.
Houetohossou et al. [24]	To systematically review deep learning and AI methods for detecting and classifying biotic (diseases, pests) and abiotic (nutrient deficiency, water, light, heavy metals) stresses in fruits and vegetables (132 papers, 2003–2022).	CNN architectures (GoogleNet, ResNet50, VGG16) dominate; ResNet50 and AlexNet are the best-performing models (>90% accuracy); fungal diseases are the most studied; apple and tomato are the most investigated crops; PlantVillage is the most used dataset (28%); major gaps: lack of field-representative data, class imbalance, limited abiotic stress research, no climate change stress prediction, and difficulty detecting multiple diseases on same leaf; West Africa is underrepresented in data collection.	This is a stress/disease-specific SLR focused on plant pathology, not fruit quality assessment or variety classification; it does not address Latin America vs. international research, lacks analysis of sensor technologies (hyperspectral, NIR, 3D) for quality grading, does not cover inclusivity, sustainability, human-centered design, AI-assisted decision support, or the research–practice gap; does not examine post-harvest vs. pre-harvest economic contexts or technology democratization as our scoping review does.
Espinoza et al. [25]	To conduct a systematic literature review (111 studies, 2020–2023) on deep learning models for fruit image analysis (classification, detection, segmentation), identifying models, datasets, metrics, challenges, and future directions.	Classical CNNs (58%) and R-CNN/YOLO (41%) dominate; Visual Transformers are rarely used (1%); apples (12%), tomatoes (8%), and bananas (6%) are most studied; controlled environments (56%) vs. natural (44%); precision, recall, F1, and mAP are key metrics; major challenges: data scarcity, labeling complexity, shape/color variability, occlusion, and computational efficiency; public datasets include Fruit 360, FruitNet, VegFru, etc.	This is a general SLR on fruit image analysis with DL without regional focus; it does not compare Latin America vs. international research, lacks analysis of sensor technologies (hyperspectral, NIR, 3D) beyond RGB/RGB-D, does not address inclusivity, sustainability, human-centered design, or AI-assisted decision support; does not examine post-harvest vs. pre-harvest economic contexts, technology democratization, or the research–practice gap as our scoping review does.
Indira et al. [26]	To present a review of fruit recognition and feature evaluation using CNN, covering fruit detection, classification, CNN architecture, and implementation considerations (MATLAB, Google Colab).	CNN (especially Faster R-CNN) is effective for fruit detection and classification; key CNN components include convolution, ReLU activation, pooling, and fully connected layers; AlexNet (2012) was a major breakthrough; Fruit 360 dataset from Kaggle is widely used; future work includes implementing and developing a framework for fruit recognition.	This is a very short, superficial review paper (6 pages) without systematic methodology; it does not compare Latin America vs. international research, lacks analysis of sensor technologies (hyperspectral, NIR, 3D), does not address inclusivity, sustainability, human-centered design, or AI-assisted decision support; does not cover post-harvest vs. pre-harvest economic contexts, technology democratization, or the research–practice gap; lacks depth and critical analysis compared to our scoping review.

2.1. Similarities and Differences with Our Scoping Review

While previous reviews have provided valuable, albeit fragmented, insights into specific facets of the field, they collectively fail to offer a holistic, critically comparative, and contextually aware synthesis. For instance, general reviews like those by Ukwuoma et al. [18] and Sawant et al. [20] have comprehensively cataloged deep learning techniques and their impres-

sive performance, often exceeding 99% accuracy on curated datasets. However, these works are largely techno-centric, focusing on algorithmic proficiency without systematic examination of the geographical, economic, or human factors that shape technological adoption. Similarly, surveys on specific applications, such as ripeness classification [22] or citrus diseases [21], offer deep dives into narrow sub-domains but lack the breadth to identify cross-cutting blind spots. Furthermore, all the reviews in Table 1—including the systematic literature reviews by Rojas Santelices et al. [19] and Espinoza et al. [25]—operate without a critical regional lens, treating the global research landscape as a monolithic entity and thus failing to acknowledge the stark disparities in resources, research priorities, and implementation contexts that characterize different parts of the world.

In stark contrast, our scoping review differs by its critical, comparative framework that explicitly contrasts Latin American and international research traditions. Crucially, our scoping review adopts a broader analytical perspective that integrates technological, geographical, and implementation-oriented dimensions. Beyond cataloguing sensing technologies and AI models, we compare regional research ecosystems, highlight methodological limitations, examine deployment readiness, and discuss cross-cutting themes such as explainable decision support, inclusivity, accessibility, and environmental sustainability. By focusing on this regional comparison, our work provides a broader overview of the situation in Latin America, highlighting apparent strengths in contextual adaptation and its significant limitations. This analysis pinpoints critical areas needing further exploration, thus covering methodological gaps prevalent in the region which may help foster a more equitable and impactful global research agenda.

2.2. Emerging Physics-Based and Biosensing Technologies for Fruit Quality Inspection

Recent advances in food quality inspection have extended beyond conventional computer vision systems toward physics-based sensing technologies capable of assessing internal physicochemical properties without destructive sampling. Optical spectroscopy, fluorescence imaging, hyperspectral imaging, near-infrared (NIR) spectroscopy, surface plasmon resonance (SPR), electrochemical biosensors, and dielectric sensing have emerged as complementary approaches for evaluating fruit freshness, soluble solids content, internal bruising, microbial contamination, pesticide residues, and biochemical composition. Unlike RGB imaging, which primarily captures external appearance, these sensing technologies provide information about internal quality attributes that are often invisible to conventional cameras.

Artificial intelligence has increasingly been integrated with these sensing modalities to improve feature extraction, pattern recognition, and predictive modelling. Deep learning architectures, support vector machines, random forests, and ensemble learning algorithms have demonstrated promising performance when processing high-dimensional spectral or biosensing data. Consequently, the combination of advanced sensing technologies with AI represents an emerging direction toward comprehensive, non-destructive food quality inspection systems capable of simultaneously evaluating both external and internal fruit characteristics.

3. Methodology

This scoping review was conducted following the PRISMA-ScR (Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews) guidelines [27]. The protocol was pre-registered in the Open Science Framework (OSF) portal (available at https://osf.io/4j3sx/overview?view_only=399c419e429c49439efc92311708ec6d) on 10 May 2026. The detailed PRISMA 2020 checklist, outlining the adherence to

the reporting guidelines for this scoping review, is available as Supplementary Material (Table S1).

3.1. Rationale for Selecting a Scoping Review

This study adopted a scoping review methodology because the primary objective was to comprehensively map the current landscape of sensor technologies and artificial intelligence applied to fruit detection, classification, and quality assessment rather than to estimate the effectiveness of a single intervention. The identified literature encompasses substantial heterogeneity in sensing modalities, computational algorithms, agricultural contexts, evaluation metrics, target fruit species, and validation protocols. Such methodological diversity prevents meaningful statistical aggregation of results and therefore makes a meta-analysis inappropriate.

Unlike a conventional systematic review, which generally focuses on answering narrowly defined effectiveness questions, a scoping review enables the identification of technological trends, research gaps, emerging sensing modalities, geographical disparities, and future research opportunities across a rapidly evolving multidisciplinary field. This approach is particularly suitable for synthesizing evidence originating from computer science, agricultural engineering, robotics, food technology, and precision agriculture, where study designs and outcome measures differ considerably.

Consequently, the objective of this review is not to determine which algorithm performs best under identical experimental conditions, but rather to provide a comprehensive map of current technological developments, compare regional research trends between Latin America and the international community, identify methodological limitations, and propose future directions for more accessible, robust, and sustainable AI-enabled agricultural systems.

3.2. Databases and Search Strategy

A comprehensive literature search was conducted across Scopus and IEEE Xplore to identify peer-reviewed journal articles and conference proceedings relevant to sensor technologies and artificial intelligence for fruit detection, classification, and quality assessment published between January 2012 and May 2026. Google Scholar was used as a complementary source to capture grey literature and conference proceedings not indexed in Scopus or IEEE Xplore. All database searches were executed to ensure temporal consistency across sources.

The search strategy employed Boolean combinations of terms related to sensor technologies, artificial intelligence, and fruit quality assessment. Given the interdisciplinary nature of the topic, which spans computer science, engineering, agricultural science, and food technology, Scopus and IEEE Xplore were selected to ensure coverage of both technical and agricultural contributions. Google Scholar was used to broaden coverage and mitigate potential database bias by capturing studies indexed outside traditional disciplinary boundaries.

The search strategy incorporated terms related to sensing modalities (including computer vision, machine vision, hyperspectral imaging, near-infrared spectroscopy, and RGB imaging), computational techniques (including machine learning, deep learning, artificial neural networks, convolutional neural networks, and YOLO), and agricultural applications (including fruit detection, fruit classification, quality assessment, ripeness detection, and defect identification). Boolean operators were used to combine these terms, ensuring comprehensive retrieval of the relevant literature. All this information is presented in Table 2.

Table 2. Number of studies retrieved per database using the systematic search strategy.

Database	Search Strategy	Results
Scopus	TITLE-ABS-KEY(("fruit" OR "fruits") AND ("classification" OR "detection" OR "grading" OR "quality assessment" OR "ripeness" OR "defect detection") AND ("deep learning" OR "machine learning" OR "neural network" OR "computer vision" OR "artificial intelligence" OR "convolutional neural network" OR "CNN" OR "YOLO" OR "image processing")) AND (LIMIT-TO (DOCTYPE, "ar") OR LIMIT-TO (DOCTYPE, "cp")) AND (LIMIT-TO (LANGUAGE, "English") OR LIMIT-TO (LANGUAGE, "Spanish"))	11,059
IEEE Xplore	("fruit" OR "fruits") AND ("classification" OR "detection" OR "grading" OR "quality assessment" OR "ripeness" OR "defect detection") AND ("deep learning" OR "machine learning" OR "neural network" OR "computer vision" OR "artificial intelligence" OR "convolutional neural network" OR "CNN" OR "YOLO" OR "image processing")	3251
Google Scholar	("computer vision" OR "machine vision" OR "artificial intelligence" OR "deep learning" OR "machine learning") AND ("fruit" OR "fruits" OR "produce") AND ("quality" OR "grading" OR "sorting" OR "detection" OR "classification" OR "ripeness")	228,000
Total		242,310

Given the extremely large volume of results retrieved from all databases—especially from Google Scholar, which returned 228,000 records—a pragmatic and targeted screening strategy was adopted, consistent with previously published scoping reviews facing similar retrieval volumes [28,29]. Results from each database were sorted by relevance, and the first 300 records from each source were screened based on title and abstract.

This threshold was chosen for three primary reasons.

- First, methodological guidance for scoping reviews acknowledges that comprehensive screening of all retrieved records is often impractical and that feasibility-based strategies are both legitimate and widely adopted when clearly reported [28,30]. Manual screening of 242,310 records would be prohibitively resource-intensive and would exceed the capacity of a scoping review, which is designed to map the literature rather than exhaustively synthesize all available evidence.
- Second, relevance ranking in academic databases—particularly in Scopus and Google Scholar, which employ citation-based algorithms—typically places the most pertinent studies at the top of search results [31]. The methodological literature on systematic searching has demonstrated that literature searching is subject to the law of diminishing returns; after a certain threshold, further searching yields increasingly fewer relevant papers [32,33]. Consequently, there comes a point where the rewards of further searching may not be worth the effort required [34].
- Third, this approach allows for a manageable yet sufficiently representative sample to understand the broad technological trends and regional differences in the development of fruit detection technologies, both in Latin America and globally. As Pham et al. [29] observed in their analysis of 272 scoping reviews, substantial variability exists in how researchers handle large retrieval volumes, and pragmatic screening strategies are commonplace in the methodology.

Although combining three complementary databases reduced the risk of selection bias, screening only the first 300 relevance-ranked records from each source may have excluded relevant studies appearing beyond this threshold. Consequently, the findings presented in this review should be interpreted as a synthesis of the most accessible and highly ranked evidence rather than an exhaustive representation of all published research. This limitation should be considered when interpreting both the regional comparisons and the identified research gaps.

The search was restricted to peer-reviewed journal articles and conference proceedings published in English or Spanish, ensuring accessibility and quality standards. Conference proceedings were included given the rapid pace of technological development in this field, where significant advances often appear first in conference venues.

During the screening process, studies were retained if they presented empirical evaluations of sensor-based systems for fruit detection, classification, or quality assessment; proposed novel computational approaches for fruit-related agricultural applications; or provided comparative analyses of different sensing or algorithmic approaches. Conceptual papers without empirical validation, purely speculative works, and studies focusing on non-agricultural applications were excluded.

The search strategy was designed to balance comprehensiveness with precision, ensuring the retrieval of both foundational studies in classical computer vision and emerging research in deep learning and edge computing. This approach allowed the review to capture the full spectrum of technological development in fruit detection, from accessible low-cost systems to sophisticated multi-modal sensing platforms, providing a solid foundation for a comparative analysis between Latin American and international research traditions.

3.3. Inclusion and Exclusion Criteria

Studies were included if they met the following criteria:

- Focused on fruit detection, classification, quality assessment, or grading using sensor technologies or artificial intelligence;
- Implemented, evaluated, or proposed computer vision, machine learning, deep learning, or sensor-based approaches for fruit-related agricultural applications;
- Involved empirical evaluation of sensor-based systems for fruit quality assessment, including defect detection, ripeness estimation, size and mass estimation, or shelf life prediction;
- Addressed fruit species of agricultural or commercial significance, including but not limited to apples, bananas, mangoes, tomatoes, blueberries, strawberries, oranges, pineapples, passion fruit, chili peppers, olives, pomegranates, and dates;
- Peer-reviewed journal articles or conference proceedings;
- Published in English or Spanish;
- Published between January 2012 and May 2026.

Exclusion criteria were as follows:

- Studies focusing exclusively on non-agricultural applications or non-fruit produce;
- Purely conceptual literature reviews or speculative works lacking sufficient methodological or design detail, including those without empirical validation;
- Studies addressing general computer vision or machine learning techniques without specific application to fruit detection or quality assessment;
- Studies focusing exclusively on pre-harvest cultivation practices (e.g., irrigation, fertilization) without sensor-based quality assessment components;
- Duplicate publications;
- Full-text unavailable;
- Studies published in languages other than English or Spanish.

Conceptual and design-oriented studies that provided detailed descriptions of sensor-based systems, computational frameworks, or proposed interventions relevant to fruit quality assessment were included only if they offered sufficient technical detail to enable replication or critical evaluation. Purely speculative works lacking methodological detail were excluded.

Conference proceedings were included given the rapid pace of technological development in this field, where significant advances in computer vision, deep learning architectures, and sensor technologies often appear first in conference venues. This inclusion criterion ensures the review captures the most current developments in fruit detection technologies.

No predefined quota was established for the inclusion of Latin American or international studies. Instead, geographical representation naturally emerged from the application of the predefined search strategy and eligibility criteria. This approach minimizes selection bias and ensures that the final evidence synthesis accurately reflects the current global research landscape. Consequently, the observed predominance of international publications should be interpreted as an indicator of existing research activity rather than as an artifact of the review methodology.

3.4. Study Selection and Quality Appraisal

The screening process was conducted by two independent reviewers (F.P.-B. and J.G.-C.) to ensure methodological rigor and minimize selection bias. Titles and abstracts of all retrieved records were initially screened against the pre-established inclusion and exclusion criteria. Subsequently, full-text articles deemed potentially relevant were retrieved and independently assessed for eligibility. Any disagreements between the two reviewers at either stage were resolved through discussion and consensus; in instances where consensus could not be reached, a third reviewer (J.J.-G.) was consulted to arbitrate the final decision. This process was designed to enhance the reliability and reproducibility of the study selection decisions.

Figure 1 presents the PRISMA flowchart documenting the study selection process. From the initial 242,310 records identified through database searching (Scopus: 11,059; IEEE Xplore: 3251; Google Scholar: 228,000), a structured screening procedure was applied. As detailed in Section 3.2, results from each database were sorted by relevance, and the first 300 records from each source were screened, resulting in 900 records being examined for potential inclusion. After the initial screening and the removal of duplicate records, a total of 387 full-text articles were assessed for eligibility.

Regarding record management and deduplication, all records retrieved from Scopus and IEEE Xplore were exported in BibTeX format and imported into Zotero (version 6.0.27) for management. The 300 records screened from Google Scholar were manually entered into the same Zotero library, with bibliographic information recorded for each potentially relevant record. Automated deduplication was performed using Zotero's "Find Duplicates" function, which compares records across title, author, year, and DOI fields. Following automated deduplication, two reviewers (F.P.-B. and J.G.-C.) independently performed manual duplicate checks to identify remaining duplicates not captured algorithmically, such as records with minor title variations or identical content published in both journal and conference formats. Discrepancies in duplicate identification were resolved through consensus.

Full-text review led to the exclusion of 352 articles for the following reasons: irrelevant population ($n = 67$), irrelevant intervention ($n = 89$), conceptual only without empirical data ($n = 104$), language other than English or Spanish ($n = 31$), full-text unavailable ($n = 18$), and duplicate publications not identified in the initial automated deduplication ($n = 43$). This rigorous selection process resulted in 35 studies that met all inclusion criteria and were included in the final synthesis.

To further enhance the robustness of our scoping review, a formal quality appraisal of the 35 included studies was conducted using an adapted version of the Mixed Methods Appraisal Tool (MMAT). Two independent reviewers (F.P.-B. and J.G.-C.) assessed each study against five criteria: (1) clarity of research objectives, (2) appropriateness of study design, (3) adequacy of participant or sample characterization, (4) validity of measurement instruments and gold standards, and (5) robustness of analysis and statistical validation. Discrepancies were resolved through consensus. Studies were rated as having "high", "moderate", or "low" methodological quality. No studies were excluded based on quality;

however, this appraisal informed the interpretative weight assigned to findings, with claims from low-quality studies being treated as preliminary and framed tentatively.

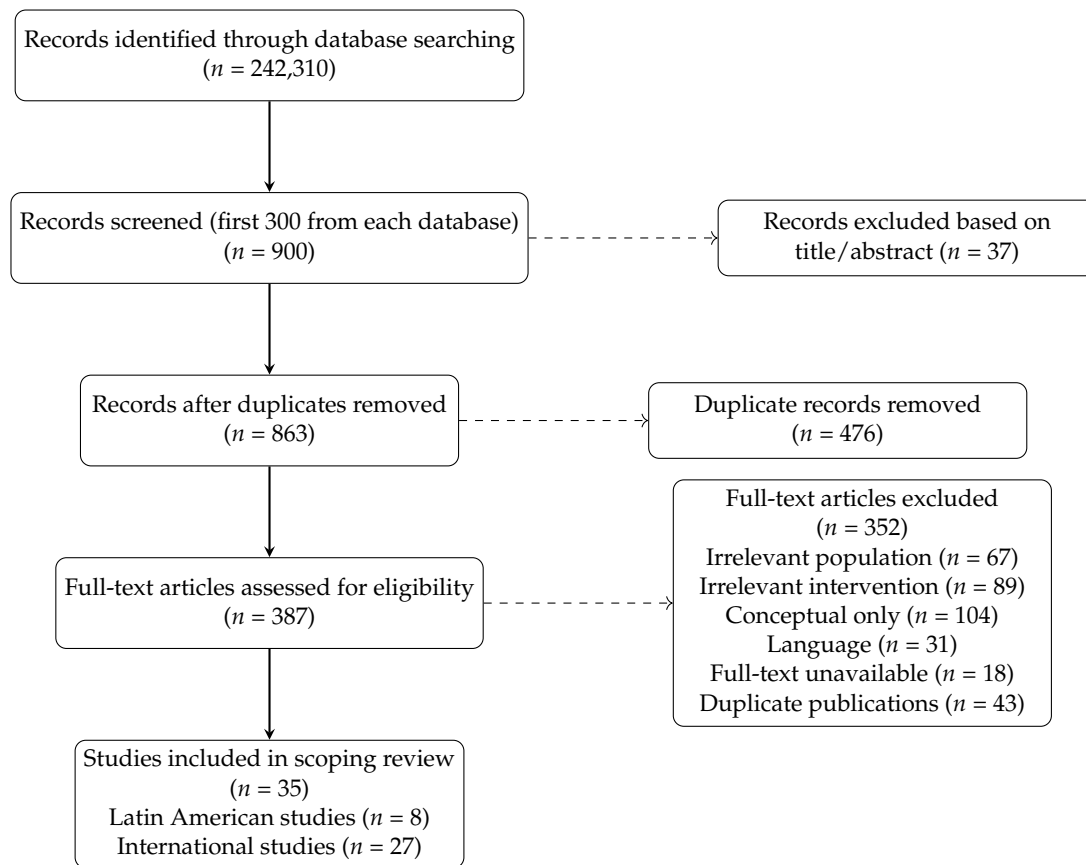


Figure 1. PRISMA flowchart of study selection.

Inter-rater reliability was formally assessed for both the study selection process and the data extraction phase. For study selection, agreement between the two independent reviewers on full-text eligibility was calculated using Cohen's kappa coefficient, yielding a value of $\kappa = 0.87$, indicating a high level of agreement. For the data extraction phase, reliability was assessed on a randomly selected subset of 30% of the included studies ($n = 11$). Agreement on extracted variables, particularly on the coding of geographical origin, technological approaches, target crops, and reported performance metrics, was calculated, resulting in a pooled agreement rate of 94.2%. These reliability checks confirm the robustness of the screening and extraction processes and minimize the risk of individual bias influencing the synthesis. Finally, although only 35 studies were ultimately included, this outcome reflects the rigorous application of predefined eligibility criteria rather than a limitation of the review. Large initial retrieval volumes are common in scoping reviews, where broad search strategies maximize sensitivity before systematic screening identifies the most relevant evidence.

To address heterogeneity in the evidence base and to facilitate the comparative analysis between Latin American and international research, studies were explicitly classified into two geographical categories.

- Latin American studies: studies originating from Ecuador, Colombia, Mexico, Peru, Argentina, and other Latin American countries ($n = 8$);
- International studies: studies originating from China, the United States, Spain, Thailand, Iran, Saudi Arabia, India, Belgium, and other non-Latin American countries ($n = 27$).

This classification was used in subsequent analysis to enable systematic comparison of technological approaches, resource utilization, and research priorities across different geographical and economic contexts.

Although one of the objectives of this review was to compare Latin American and international research, the review did not attempt to artificially equalize the number of studies between both groups. Instead, the comparison was based on the best available evidence identified through the systematic search process. This strategy preserves methodological transparency while allowing meaningful qualitative comparisons regarding technological maturity, sensing modalities, computational approaches, and implementation contexts.

3.5. Quality Appraisal Results

The methodological quality of the 35 included studies, assessed using the adapted Mixed Methods Appraisal Tool (MMAT) criteria, is summarized in Table 3. Two independent reviewers (F.P.-B. and J.G.-C.) evaluated each study against five criteria: (1) clarity of research objectives, (2) appropriateness of study design, (3) adequacy of sample characterization, (4) validity of measurement instruments, and (5) robustness of analysis. Overall, 27 studies (77.1%) were rated as high quality, 6 studies (17.1%) as moderate quality, and 2 studies (5.7%) as low quality. The primary reasons for lower quality ratings were limited sample sizes ($n < 100$ images), absence of external validation, and insufficient reporting of gold standard procedures.

Table 3. Summary of Methodological Quality Assessment (MMAT) by Geographic Region.

Region	High Quality	Moderate Quality	Low Quality	Total
Latin American Studies	4 (50.0%)	3 (37.5%)	1 (12.5%)	8
International Studies	23 (85.2%)	3 (11.1%)	1 (3.7%)	27
Total	27 (77.1%)	6 (17.1%)	2 (5.7%)	35

The lower proportion of high-quality studies in the Latin American corpus (50.0%) compared to international research (85.2%) reflects differences in available resources, dataset sizes, and validation infrastructure. However, it should be noted that low-quality studies were not excluded from the synthesis; rather, their findings were interpreted with appropriate caution, with claims from low-quality studies being treated as preliminary and framed tentatively. This approach preserves the breadth of evidence while enabling readers to weigh the strength of conclusions appropriately.

All in all, the quality appraisal results informed the interpretative weight assigned to findings throughout this review. Specifically, claims from high-quality studies were treated as robust evidence, while findings from moderate- and low-quality studies were presented with appropriate caveats. No study was excluded based on quality, as the scoping review methodology prioritizes breadth of evidence over critical appraisal thresholds. Nevertheless, readers should interpret the synthesis with awareness of the methodological heterogeneity across the included studies.

3.6. Data Extraction

Data extraction was performed by one reviewer (F.P.-B.) using a structured coding framework developed a priori to ensure consistency across studies. The extraction form was piloted on three randomly selected studies and refined to capture all relevant variables. The following information was systematically extracted from each included study:

- Authors and year of publication;
- Country and geographical region of origin (Latin America vs. international);

- Study type (empirical vs. conceptual/design-oriented);
- Target fruit species and varieties;
- Dataset characteristics (sample size, image resolution, collection conditions);
- Sensing modalities employed (RGB imaging, hyperspectral, NIR spectroscopy, 3D scanning, X-ray, multi-spectral);
- Computational techniques and algorithms (classical image processing, machine learning, deep learning architectures);
- Gold standard or reference method used for validation;
- Reported performance metrics (accuracy, precision, recall, F1-score, mAP, RMSE, etc.);
- Stage of agricultural production (pre-harvest, post-harvest, or both);
- Application context (quality grading, defect detection, ripeness assessment, size/mass estimation, shelf life prediction, robotic harvesting);
- Key findings and contributions;
- Limitations and challenges reported by authors.

Although funding source was not established as a primary extraction variable during protocol development, funding acknowledgements were qualitatively examined whenever reported by the original studies. This complementary analysis was intended to provide contextual information regarding the research environments supporting AI-enabled fruit detection systems and to facilitate interpretation of regional differences in technological development.

In addition to image-based sensing modalities, particular attention was given during data extraction to studies employing physics-based sensing technologies, including hyperspectral imaging, near-infrared spectroscopy, X-ray imaging, fluorescence sensing, dielectric sensing, and biosensor-based approaches whenever these technologies were integrated with artificial intelligence for fruit quality inspection. Although comparatively fewer studies employed these advanced sensing techniques, they represent an important emerging research direction for non-destructive agricultural quality assessment.

To ensure accuracy and consistency, the extracted data were independently verified by a second reviewer (J.G.-C.). Any discrepancies between the two reviewers were discussed and resolved through consensus, with a third reviewer (J.J.-G.) consulted when necessary.

Inter-rater reliability for the data extraction phase was formally assessed on a randomly selected subset of 30% of the included studies ($n = 11$). Agreement on extracted variables, particularly on the coding of geographical origin, technological approaches, target crops, and reported performance metrics, was calculated, resulting in a pooled agreement rate of 94.2%. These reliability checks confirm the robustness of the extraction process and minimize the risk of individual bias influencing the synthesis.

To address heterogeneity in the evidence base and to facilitate meaningful synthesis, studies were explicitly classified into two categories:

- Empirical studies: involving implementation and/or evaluation with reported data from experiments, field trials, or real-world validation ($n = 32$);
- Conceptual/design-oriented studies: proposing or describing systems, frameworks, or interventions without empirical validation or with limited preliminary testing ($n = 3$).

This distinction was used in subsequent analysis to differentiate between validated evidence and emerging design insights, ensuring that conclusions drawn from empirical studies were given appropriate weight compared to conceptual proposals.

Additionally, studies were classified according to their geographical origin to enable the comparative analysis between Latin American and international research traditions:

- Latin American studies: originating from Ecuador, Colombia, Mexico, Peru, Argentina, and other Latin American countries ($n = 8$);

- International studies: originating from China, the United States, Spain, Thailand, Iran, Saudi Arabia, India, Belgium, and other non-Latin American countries ($n = 27$).

This geographical classification, combined with the systematic extraction of technological and methodological characteristics, provided the foundation for the thematic analysis organized around the research questions, enabling a nuanced comparison of how different contexts, resources, and priorities shape the development of fruit detection technologies across the globe.

4. Results

The 35 included studies are presented in Tables A1 and A2. They span diverse geographical contexts, technological approaches, and agricultural applications, revealing a rich and multifaceted research landscape. In this section, we synthesize the evidence across five thematic dimensions: sensor technologies, computational approaches, target crops, performance outcomes, and stage of agricultural production, followed by an examination of the research-to-practice gap.

4.1. Sensor Technologies

As synthesized in Table 4, the included studies employ a remarkably diverse array of sensing technologies and computational techniques. At the most fundamental level, conventional RGB imaging remains the predominant sensing modality, appearing in 34 of the 35 studies (97%) due to its accessibility, low cost, and proven effectiveness for external quality assessment tasks. However, a significant subset of studies incorporates more advanced sensing technologies, including near-infrared spectroscopy, hyperspectral imaging, multispectral imaging, 3D infrared scanning, X-ray radiography, and multi-spectral filter systems. This technological progression from simple RGB cameras to sophisticated multi-modal sensing systems reflects the field's evolution toward comprehensive quality assessment that captures both external appearance and internal properties.

The regional analysis suggests differences in the technological approaches adopted by the included studies. The selected Latin American studies rely almost exclusively on RGB/Color Imaging, with all 8 studies employing this modality, complemented only by isolated uses of ultrasonic/sonic sensors and other technologies. This concentration reflects both resource constraints and a pragmatic focus on developing accessible solutions using widely available hardware. By contrast, among the international studies included in this review, a greater diversity of sensing technologies was observed, with 26 studies employing RGB imaging alongside significant adoption of advanced sensing technologies: hyperspectral imaging (5 studies), multispectral imaging (5 studies), Vis/NIR spectroscopy (5 studies), 3D/depth sensors (4 studies), multi-sensor fusion (4 studies), and various other modalities including thermal imaging, X-ray imaging, acoustic/vibration sensors, infrared imaging, and gas/ethylene sensors. However, this finding should be interpreted cautiously because the international corpus encompasses highly heterogeneous research environments and should not be viewed as representing a single, unified technological tradition.

Table 4. Distribution of Sensor Technologies Used in Fruit Quality Assessment Studies by Geographic Region.

Region	Technology	Studies	Count
Latin American	RGB/Color Imaging	Gonzalez et al. [35], Larcher et al. [36], Nieto and Rangel [37], Silva and Lizcano [38], Sucari et al. [39], Alvarado and Molina [40], Constante et al. [41], Cruz-Domínguez et al. [42]	8
	Ultrasonic/Sonic Sensors	Cruz-Domínguez et al. [42]	1
	Other Technologies	Larcher et al. [36]	1
International	RGB/Color Imaging	Ponce et al. [43], Shi et al. [44,45], Sukkuea et al. [46], Tempelaere et al. [47], Thinh et al. [48], Wakhare et al. [49], Wang et al. [50], Wanthong et al. [51], Yan et al. [52], Yu et al. [53], Zhang et al. [54,55], Zhao et al. [56], Aldakhil and Almutairi [57], Apostolopoulos et al. [58], Dutta et al. [59], Esfandiari Fard et al. [60], Fan et al. [61,62], Hosainpour et al. [63], Hu et al. [64], Kevrešan et al. [65], Moallem et al. [66], Mputu et al. [67], Ni et al. [68]	26
	Hyperspectral Imaging	Tempelaere et al. [47], Zhang et al. [54], Fan et al. [61], Hu et al. [64], Kevrešan et al. [65]	5
	Multispectral Imaging	Zhang et al. [54], Aldakhil and Almutairi [57], Fan et al. [61], Kevrešan et al. [65], Mputu et al. [67]	5
	Vis/NIR Spectroscopy	Zhang et al. [54], Zhao et al. [56], Hosainpour et al. [63], Hu et al. [64], Srinivasagan et al. [69]	5
	Thermal Imaging	Fan et al. [61], Hu et al. [64], Kevrešan et al. [65]	3
	Gas/Ethylene Sensors	Apostolopoulos et al. [58], Kevrešan et al. [65], Srinivasagan et al. [69]	3
	3D/Depth Sensors	Sukkuea et al. [46], Thinh et al. [48], Wang et al. [50], Hu et al. [64]	4
	X-ray Imaging	Tempelaere et al. [47], Zhang et al. [55], Hosainpour et al. [63]	3
	Acoustic/Vibration Sensors	Fan et al. [62], Kevrešan et al. [65], Moallem et al. [66]	3
	Infrared Imaging	Yan et al. [52], Fan et al. [61], Hu et al. [64]	3
	Multi-Sensor Fusion	Ponce et al. [43], Zhao et al. [56], Esfandiari Fard et al. [60], Kevrešan et al. [65]	4
	Other Technologies	Hosainpour et al. [63], Kevrešan et al. [65]	2

Although RGB imaging remains the dominant sensing modality among the reviewed studies, recent studies suggest a growing transition toward complementary physics-based sensing technologies capable of capturing internal fruit characteristics. Near-infrared spectroscopy enables estimation of soluble solids content and internal maturity, hyperspectral imaging facilitates simultaneous spatial and spectral analysis, while X-ray imaging allows non-destructive identification of internal defects. Emerging biosensing techniques—including electrochemical biosensors and SPR-based sensing—although still underrepresented in the reviewed literature, may offer considerable potential for future AI-enabled food quality inspection by providing rapid biochemical characterization of freshness, contamination, and quality degradation.

4.2. Computational Approaches

The computational approaches of all selected studies are synthesized in Table 5. They exhibit similar diversity, ranging from classical machine learning techniques to state-of-the-art deep learning architectures. Traditional computer vision methods—including Otsu thresholding, edge detection, color space transformations, and handcrafted feature extraction—remain prevalent, particularly in Latin American studies where computational resources may be more limited. Indeed, among the included Latin American studies, a greater reliance was observed on classical machine learning and computer vision techniques, with Classical ML/CV appearing in 4 studies, Artificial Neural Networks (ANNs) in 3 studies, and Support Vector Machines (SVM) in 2 studies. This preference may reflect differences in available resources and a pragmatic focus on developing accessible, deployable solutions using well-established methods that require less computational infrastructure. In contrast, the included international studies more frequently employed substantially greater adoption of state-of-the-art deep learning architectures, with Convolutional Neural Networks (CNNs) appearing in 10 studies, YOLO variants in 7 studies, Vision Transformers (ViTs) in 6 studies, and Transformer-based DETR series in 3 studies. This observed difference may reflect the uneven access to advanced computational resources and large-scale datasets across different research contexts.

Table 5. Distribution of Computational Approaches Used in Fruit Quality Assessment Studies by Geographic Region.

Region	Approach	Studies	Count
Latin American	Classical ML/Computer Vision	Gonzalez et al. [35], Larcher et al. [36], Silva and Lizcano [38], Cruz-Domínguez et al. [42]	4
	Convolutional Neural Networks (CNNs)	Alvarado and Molina [40], Constante et al. [41]	2
	Artificial Neural Networks (ANNs)	Gonzalez et al. [35], Constante et al. [41], Cruz-Domínguez et al. [42]	3
	Support Vector Machines (SVM)	Gonzalez et al. [35], Silva and Lizcano [38]	2
	Clustering (K-means, MBSAS)	Silva and Lizcano [38]	1
	Principal Component Analysis (PCA)	Gonzalez et al. [35]	1
	Linear Discriminant Analysis (LDA)	Gonzalez et al. [35], Silva and Lizcano [38]	2
International	Classical ML/Computer Vision	Thinh et al. [48], Zhang et al. [54], Hosainpour et al. [63], Kevrešan et al. [65], Moallem et al. [66], Ni et al. [68]	6
	Convolutional Neural Networks (CNNs)	Tempelaere et al. [47], Zhang et al. [55], Zhao et al. [56], Aldakhil and Almutairi [57], Apostolopoulos et al. [58], Dutta et al. [59], Fan et al. [61,62], Mputu et al. [67], Ni et al. [68]	10
	YOLO Variants	Shi et al. [44,45], Wakhare et al. [49], Wang et al. [50], Yan et al. [52], Yu et al. [53], Fan et al. [62]	7
	Vision Transformers (ViTs)	Shi et al. [45], Yan et al. [52], Aldakhil and Almutairi [57], Apostolopoulos et al. [58], Dutta et al. [59], Esfandiari Fard et al. [60]	6
	Hybrid (DL Feature Extraction + ML Classifiers)	Shi et al. [44], Zhao et al. [56], Apostolopoulos et al. [58], Mputu et al. [67]	4
	Transformer-based (DETR series)	Shi et al. [44,45], Yan et al. [52]	3
	Artificial Neural Networks (ANNs)	Thinh et al. [48], Hosainpour et al. [63], Ni et al. [68]	3
	Support Vector Machines (SVM)	Fan et al. [61], Hosainpour et al. [63], Moallem et al. [66], Mputu et al. [67]	4
	Random Forest / Ensemble	Wakhare et al. [49], Hosainpour et al. [63], Mputu et al. [67], Srinivasagan et al. [69]	4
	Knowledge Distillation	Shi et al. [44,45]	2
	Multi-Task Learning	Aldakhil and Almutairi [57], Dutta et al. [59], Esfandiari Fard et al. [60]	3
	Instance Segmentation	Sukkuea et al. [46], Wang et al. [50], Yu et al. [53]	3
	Principal Component Analysis (PCA)	Zhang et al. [54], Kevrešan et al. [65]	2
	Transfer Learning	Tempelaere et al. [47], Aldakhil and Almutairi [57], Mputu et al. [67], Ni et al. [68]	4

In parallel, the adoption of deep learning has become increasingly widespread, with architectures such as Convolutional Neural Networks (CNNs) in various configurations, You Only Look Once (YOLO) variants for real-time detection, Vision Transformers (ViTs), and specialized architectures like EMBS-DETR and BR-DETR-Prune demonstrating exceptional performance. Hybrid approaches that combine deep learning for feature extraction with traditional classifiers, as exemplified by the CNN-SVM methodology employed by Mputu et al. [67], represent a particularly elegant synthesis of both traditions. The international research landscape also suggests the emergence of more sophisticated techniques, including Knowledge Distillation (2 studies), Multi-Task Learning (3 studies), Instance Segmentation (3 studies), and Transfer Learning (4 studies), which are entirely absent from the Latin American corpus. Furthermore, classical techniques such as Principal Component Analysis (PCA) appear in both regions, though more frequently in international studies, while Linear Discriminant Analysis (LDA) and Clustering methods remain primarily confined to Latin American research. This distribution suggests that the included international studies increasingly focused on pushing the boundaries of what is technologically possible, developing novel architectures and training paradigms, while Latin American research continues to excel in the thoughtful application of established techniques to solve practical agricultural challenges.

4.3. Target Crops and Geographical Research Priorities

The target crops examined across the 35 studies, as presented in Table 6, reveal important patterns about research priorities and economic contexts. Latin American research indicates a strong focus on export-oriented fruits of regional economic significance, including bananas, pineapples, passion fruit, chili peppers, granadilla, strawberries, and dried

figs. This emphasis reflects the critical importance of agricultural exports to Latin American economies and the pressing need for quality control technologies that can enhance international competitiveness. Notably, the crops studied in Latin America—such as passion fruit in Colombia [35], pineapples in Colombia [38], chili peppers in Mexico [42], and bananas in Panama [37]—are directly tied to the region’s agricultural export portfolio, demonstrating a tight coupling between research agendas and economic priorities. International research, by contrast, displays greater crop diversity, encompassing globally significant fruits such as apples (11 studies), tomatoes (5 studies), mangoes (5 studies), blueberries, oranges, pomegranates, grapes, and peaches, alongside more specialized crops like litchi, dates, olives, and white mulberry. This broader crop coverage reflects both the larger scale of international agricultural research and the presence of well-established fruit industries with substantial investments in automation. Notably, apples emerge as the most extensively studied crop, reflecting the commodity’s global economic significance.

Table 6. Distribution of Target Crops Across Studies by Geographic Region.

Region	Crop	Studies	Research Location	Research Focus
Latin American	Banana	Nieto and Rangel [37]	Panama	Ripeness classification, shelf-life estimation
	Passion Fruit	Gonzalez et al. [35]	Colombia	Mass and volume estimation
	Chili Pepper	Cruz-Domínguez et al. [42]	Mexico	Size and color classification
	Pineapple	Silva and Lizcano [38]	Colombia	Maturity assessment
	Granadilla	Nieto and Rangel [37]	Colombia	Maturity detection
	Strawberry	Constante et al. [41]	Ecuador	Shape and defect classification
	Dried Fig	Larcher et al. [36]	Argentina	Defect/ damage estimation
International	Apple	Tempelaere et al. [47], Zhang et al. [54,55], Zhao et al. [56], Fan et al. [61,62], Hu et al. [64], Moallem et al. [66]	Belgium, China, USA, Spain	Defect detection, bruise identification, grading, internal quality, sorting
	Tomato	Wakhare et al. [49], Yan et al. [52], Mputu et al. [67]	China, Egypt, India, Saudi Arabia	Maturity detection, disease detection, quality grading
	Mango	Thinh et al. [48], Wanthong et al. [51]	Vietnam, Thailand	Classification, size estimation, weight estimation
	Blueberry	Shi et al. [45]	China	Maturity detection, yield estimation
	Orange	Sukkuea et al. [46]	Thailand	Volume estimation, quality grading
	Peach	Fan et al. [62]	China	Fruit thinning detection
	Litchi	Wang et al. [50]	China	Picking robot, picking point localization
	Dates	Srinivasagan et al. [69]	Saudi Arabia	Shelf-life estimation, freshness monitoring
	Olive	Ponce et al. [43]	Spain	Size and mass estimation, grading
	Strawberry	Yu et al. [53]	China	Picking point localization, harvesting robot
	White Mulberry	Hosainpour et al. [63]	Iran	Quality grading
	Apple, Banana, Orange, Pomegranate, Lime, Guava	Aldakhil and Almutairi [57], Dutta et al. [59]	Saudi Arabia, India (dataset-based)	Multi-fruit classification and quality grading
	Multiple fruits (22 types)	Apostolopoulos et al. [58], Esfandiari Fard et al. [60]	International (USA)	Freshness detection, type classification

The geographical distribution of research also reveals important disparities in research capacity and infrastructure. Latin American studies predominantly focus on single crops, with limited multi-crop investigations, while international research increasingly addresses multiple fruit types simultaneously, leveraging large-scale public datasets such as Fruit-Net [57] and the comprehensive Freshness44 collection comprising 53,616 images across 22 fruit and vegetable types [60]. This divergence suggests that international research benefits from more extensive data infrastructure, enabling broader generalization studies that are less feasible in resource-constrained Latin American contexts. However, the focused crop specialization in Latin American research also enables deeper investigation of region-

ally specific quality challenges, potentially yielding more directly applicable solutions for local agricultural industries.

4.4. Performance Outcomes and Generalization Challenges

As shown in Table 7, the performance results reported across the studies demonstrate consistently high accuracy levels, with the vast majority of systems achieving classification rates exceeding 90% and many approaching or surpassing 99% accuracy. Particularly noteworthy are the achievements of deep learning-based systems, which routinely outperform classical approaches by substantial margins. For instance, Esfandiari Fard et al. [60] achieved 99.65% freshness detection accuracy using Multi-Task NoisyViT, while Aldakhil and Almutairi [57] reported 99.49% fruit classification accuracy with EfficientNetV2. Even more remarkably, Hosainpour et al. [63] achieved perfect 100% classification accuracy for dried white mulberry quality grading using a 23-14-2 ANN structure, demonstrating that simpler approaches can achieve exceptional performance when appropriately designed for specific applications.

Table 7. Summary of Performance Trends and Generalization Capabilities by Region.

Aspect	Latin American Studies	International Studies
Typical Accuracy Range	80–96%	90–99.65%
Maximum Reported Accuracy	96.36% (Silva and Lizcano [38])	100% (Hosainpour et al. [63]); 99.65% (Esfandiari Fard et al. [60])
Performance Variability	Moderate	High (82% to 100%)
Generalization Testing	Limited; often single dataset	Extensive; multiple test sets, cross-validation, public datasets
Key Challenge: Dataset Shift	Limited exploration	Documented performance drop across conditions (e.g., Tempelaere et al. [47]: 100% → 82% across different orchards and storage conditions)
Validation Rigor	Typically single-condition validation	Multi-condition testing (different environments, lighting, varieties, storage histories)
Real-World Applicability	Promising but requires further validation	Demonstrated but with notable generalization gaps

The regional comparison reveals notable differences in both the magnitude and variability of reported performance. Latin American studies typically report accuracy ranges between 80% and 96%, with the maximum reported accuracy of 96.36% achieved by Silva and Lizcano [38] for pineapple maturity classification, validated against three expert evaluators with over 10 years of experience. This relatively narrower performance range reflects the predominant use of classical machine learning techniques and smaller, more controlled datasets. International studies, by contrast, exhibit both higher peak performance and substantially greater variability, with reported accuracies ranging from 82% to 100%. This broader range reflects the diversity of approaches—from simple ANNs achieving perfect classification on specific tasks to sophisticated deep learning architectures tested across more challenging, heterogeneous datasets.

However, these impressive accuracy figures must be interpreted with appropriate caution. Many studies report performance on carefully curated datasets that may not fully represent the variability encountered in real agricultural settings. The work of Tempelaere et al. [47] is particularly instructive in this regard, demonstrating that while GoogLeNet achieved 100% accuracy on one test dataset, performance dropped to 82% on a different dataset from varying conditions. This finding highlights the critical challenge of generalization and the importance of rigorous validation across diverse environments, lighting conditions, fruit varieties, and storage histories.

The disparity in generalization testing between regions is equally telling. Latin American studies typically employ single-condition validation with limited testing across different environments, whereas international studies increasingly adopt multi-condition testing protocols involving different environments, lighting conditions, fruit varieties, and storage histories. This difference is exemplified by Tempelaere et al. [47], who tested their model across multiple datasets from different orchards and storage conditions, revealing the significant performance degradation that can occur when models are deployed outside their training distribution. The use of cross-validation, multiple test sets, and public datasets in international research provides stronger evidence of generalizability, yet also exposes the persistent challenge of dataset shift that Tempelaere et al. [47] documented. This suggests that while international research may achieve higher peak performance, this performance may be more fragile under real-world conditions, whereas the more modest but consistent performance of Latin American systems, validated under more controlled conditions, may translate more reliably to specific local agricultural contexts.

4.5. Stage of Agricultural Production

An additional dimension of analysis concerns the stage of agricultural production addressed by each study. According to Table 8, the majority of research focuses on post-harvest applications, including quality grading, defect detection, ripening assessment, and shelf life prediction. This emphasis reflects the immediate commercial value of quality control technologies for packing houses, processing facilities, and supply chain management. The predominance of post-harvest research is particularly pronounced in Latin American studies, where all 8 studies focus exclusively on post-harvest applications, addressing critical needs for export quality control in crops such as passion fruit, pineapples, bananas, and chili peppers. This concentration reflects the urgent commercial imperative to reduce post-harvest losses and meet stringent international quality standards. However, a growing body of research addresses pre-harvest applications, particularly in international studies, with Yan et al. [52] and Fan et al. [62] developing field-based detection systems for tomato maturity and young peach detection, respectively. The integration of robotic harvesting systems, as demonstrated by Wang et al. [50] and Yu et al. [53], represents the most advanced frontier, combining detection, localization, and physical manipulation in complete agricultural automation systems.

The analysis also reveals important patterns regarding dataset characteristics. Latin American studies typically employ smaller datasets (hundreds of images) under controlled conditions, while international studies frequently employ substantially larger datasets (tens of thousands of images), enabling more robust model training and better generalization. The use of public datasets has emerged as an important strategy for enabling comparison across studies, with Esfandiari Fard et al. [60] merging five public datasets into the comprehensive Freshness44 collection of 53,616 images. This disparity has direct implications for model robustness: larger, more diverse datasets yield systems better equipped for real-world variability, while smaller, homogeneous datasets may yield systems highly effective for specific local conditions but less generalizable to other contexts. Similarly, Latin American research predominantly focuses on single crops, while international studies increasingly address multiple fruit types simultaneously, reflecting more extensive data infrastructure and collaborative networks.

Table 8. Distribution of Studies by Agricultural Production Stage and Dataset Characteristics.

Region	Production Stage	Studies	Key Applications	Dataset Scale
Latin American	Post-Harvest	Gonzalez et al. [35], Larcher et al. [36], Silva and Lizcano [38], Cruz-Domínguez et al. [42]	Quality grading, defect detection, maturity assessment	Small (hundreds of images)
	Post-Harvest	Nieto and Rangel [37]	Ripening classification, shelf-life prediction	Small (hundreds of images)
	Post-Harvest	Sucari et al. [39]	Fruit classification, pattern recognition	Small (50 records, 6 varieties)
	Post-Harvest	Alvarado and Molina [40]	Fruit classification, quality grading	Medium (13,516 images, 13 classes)
	Post-Harvest	Constante et al. [41]	Defect detection, shape classification	Small (50 images)
International	Pre-Harvest	Yan et al. [52], Fan et al. [62]	Field-based tomato maturity detection, young peach detection	Medium–Large (2327–2386 images)
	Pre-Harvest	Wang et al. [50]	Litchi picking point localization, harvesting robot	Medium (1000 images)
	Pre-Harvest	Yu et al. [53]	Strawberry picking point detection, harvesting robot	Small (2000 images)
	Post-Harvest	Tempelaere et al. [47], Zhang et al. [54,55], Zhao et al. [56], Fan et al. [61], Hu et al. [64], Moallem et al. [66]	Defect detection, bruise identification, quality grading, internal quality assessment	Small–Medium (100–300 samples)
	Post-Harvest	Shi et al. [45], Sukkuea et al. [46], Thinh et al. [48], Wanthong et al. [51], Hosainpour et al. [63], Mputu et al. [67]	Size estimation, quality grading, maturity assessment, freshness detection	Medium–Large (200–2400 images)
	Post-Harvest	Ponce et al. [43], Aldakhil and Almutairi [57], Apostolopoulos et al. [58], Dutta et al. [59], Esfandiari Fard et al. [60], Srinivasagan et al. [69]	Quality classification, freshness detection, shelf-life estimation, multi-fruit grading	Large (9000–53,616 images; public datasets)
	Post-Harvest	Shi et al. [44], Wakhare et al. [49], Ni et al. [68]	Fruit detection (YOLO), banana freshness, pomegranate disease detection	Medium (600–3976 images)

4.6. Research-to-Practice Gap and Deployment Challenges

Finally, the results reveal a notable gap between research and practice. They are synthesized in Table 9. While the reviewed studies demonstrate remarkable technical capabilities, the transition from laboratory prototypes to field-deployable systems remains challenging. Issues of cost, reliability, user training, and integration with existing agricultural workflows continue to impede widespread adoption.

Table 9. Research-to-Practice Gap and Deployment Challenges by Region.

Region	Study	Deployment Status	Key Challenges Identified	Edge/Field Deployment
Latin American	Gonzalez et al. [35]	Laboratory prototype	Dataset size, validation scope	No
	Silva and Lizcano [38]	Laboratory prototype	Controlled conditions only	No
	Cruz-Domínguez et al. [42]	Algorithm development	Imbalanced data, field validation needed	No
	Nieto and Rangel [37]	Prototype development	Small sample size, field testing needed	No
	Constante et al. [41]	Laboratory system	Real-time performance, environmental variability	No
	Alvarado and Molina [40]	Model training	Moderate accuracy, deployment not addressed	No
	Larcher et al. [36]	Algorithm proposal	Approximate results, pre-processing challenges	No
International	Srinivasagan et al. [69]	Edge-deployed system	Addressed: low-cost spectral sensor + TinyML on microcontroller	Yes (Nano 33 BLE)
	Shi et al. [45]	Edge-deployed system	Addressed: Jetson Orin Nano deployment, real-time inference (20.5 FPS)	Yes (NVIDIA Jetson)
	Wang et al. [50]	Robotic system prototype	Integration complexity, field reliability	Partial (robot prototype)
	Yu et al. [53]	Robotic system prototype	Field testing, occlusion handling, real-time constraints	Partial (robot prototype)
	Fan et al. [61]	Commercial sorting machine	Cost, illumination design, real-time performance (72 ms/apple)	Yes (4-line sorting machine)
	Zhang et al. [55]	Commercial prototype	Bruising, throughput (7.5–10.5 fruits/s), field ruggedness	Yes (harvest-assist platform)
	Zhao et al. [56]	Laboratory robot system	Processing speed (5.2 s/apple), scalability	No
	Tempelaere et al. [47]	Laboratory system	Generalization across conditions, industrial integration	No
	Fan et al. [62]	Algorithm development	Edge deployment not yet tested, hardware limitations	No
	Apostolopoulos et al. [58]	Model development	Real-world validation needed, computational demands	No

The regional analysis reveals a contrast in deployment readiness. Latin American studies remain exclusively at the laboratory prototype or algorithm development stage, with no reported field or edge deployment. Studies such as those by Gonzalez et al. [35], Silva and Lizcano [38], and Constante et al. [41] demonstrate promising laboratory results but face significant barriers including limited dataset size, controlled validation conditions, and the absence of integration with real agricultural workflows. This pattern reflects the resource constraints and infrastructure limitations that characterize research environments in the region, as well as the challenge of translating academic prototypes into commercially viable systems.

International studies, by contrast, exhibit a more diverse deployment landscape, ranging from laboratory prototypes to commercial systems. Notable examples of advanced deployment include the commercial sorting machine developed by Fan et al. [61], which operates at 5 fruits/s on a 4-line system with processing times below 72 ms per apple; the infield grading and sorting system by Zhang et al. [55], validated at throughputs of 7.5–10.5 fruits/s with sorting accuracy exceeding 99%; and the robotic harvesting systems developed by Wang et al. [50] and Yu et al. [53], which integrate detection, localization, and physical manipulation. These systems demonstrate that the transition from laboratory to field is technically feasible, yet they also reveal the substantial engineering effort and investment required to achieve operational reliability in agricultural environments.

Nevertheless, promising developments in edge computing, exemplified by Srinivasagan et al. [69] and Shi et al. [45], suggest that the gap between research and practice may narrow significantly in the coming years, particularly as affordable, energy-efficient hardware becomes increasingly capable of running sophisticated AI models directly in the field. Srinivasagan et al. [69] demonstrated that a TinyML-based shelf life estimator for date fruits could be deployed on an Arduino Nano 33 BLE Sense microcontroller, achieving real-time inference with only 1.8 KB of RAM and 10.9 KB of flash memory, while operating on a battery for up to one week. Similarly, Shi et al. [45] achieved 20.5 FPS real-time inference on an NVIDIA Jetson Orin Nano for blueberry maturity detection, demonstrating that Transformer-based architectures can be effectively deployed on edge devices through model pruning and quantization techniques. These developments are particularly significant for Latin American contexts, where the affordability and accessibility of edge computing platforms could democratize access to advanced AI capabilities without requiring substantial infrastructure investment.

5. Discussion

The application of sensor technologies and artificial intelligence for fruit detection, classification, and quality assessment represents one of the most dynamic frontiers in agricultural technology. As we examine the global landscape of research in this field, a fascinating dichotomy emerges between the approaches adopted in Latin America and those pursued in other regions of the world. This divide is not merely geographic but reflects fundamental differences in research priorities, available resources, institutional support, and the specific agricultural challenges that each region faces. Through a comprehensive analysis of the scientific literature, we can trace the contours of these two technological worlds, understanding their respective strengths, limitations, and the potential pathways toward a more integrated and impactful future. Unless otherwise indicated, the following interpretations represent the authors' synthesis of the reviewed evidence and should not be interpreted as direct findings reported by the individual primary studies.

5.1. *Why a Scoping Review Provides Added Value*

The added value of the present scoping review lies in its ability to integrate evidence originating from highly heterogeneous technological domains. Rather than ranking algorithms

according to isolated performance metrics, the review suggests broader research patterns that would remain largely invisible within a conventional effectiveness-focused systematic review. These include regional technological disparities, differences in sensing strategies, deployment readiness, accessibility of AI technologies, and underexplored dimensions such as sustainability, inclusivity, and human-centered agricultural innovation. Consequently, this review provides researchers, policymakers, and practitioners with a strategic overview of the field while simultaneously identifying priorities for future research.

5.2. *The Latin American Approach: Practical Innovation Amidst Constraints*

The research landscape in Latin America suggests a distinctive technological philosophy shaped by resource awareness and a deep commitment to solving real-world agricultural challenges. Rather than pursuing cutting-edge performance at any cost, researchers across the region have cultivated a pragmatic approach that prioritizes accessibility, affordability, and contextual relevance. This pragmatic orientation has given rise to a body of work characterized by clever adaptation of classical computer vision techniques, the creative use of low-cost hardware platforms such as Raspberry Pi and Arduino, and a persistent focus on crops of regional economic significance. While this approach may not achieve the same benchmark performance as its international counterparts, it indicates remarkable ingenuity in delivering deployable solutions that address genuine industrial needs within realistic technological and economic constraints.

It is important to emphasize that the smaller number of Latin American studies identified in this review should not be interpreted as a methodological imbalance but rather as an accurate reflection of the current state of scientific production retrieved through the predefined search strategy. This finding itself constitutes one of the principal contributions of the review, highlighting regional disparities in research capacity and technological development.

5.2.1. A Philosophy of Pragmatic Problem-Solving

Selected Latin American research in fruit detection technologies exhibits a deeply pragmatic philosophy. Researchers in this region have consistently demonstrated remarkable ingenuity in developing effective solutions despite significant resource constraints. The work emerging from countries like Ecuador, Colombia, Mexico, and Peru reflects a fundamental understanding that agricultural technology must be accessible, affordable, and directly relevant to local needs. This is not merely a constraint but has become a distinctive strength, fostering a culture of innovation that prioritizes practical deployability over technological sophistication for its own sake.

The research of [42] on dried chili pepper classification perfectly exemplifies this approach. Rather than employing the most advanced deep learning architectures, these researchers developed an effective artificial neural network system using straightforward feature extraction from 8-bit grayscale histograms. The system, while achieving what might be considered modest by international standards, directly addressed a genuine industrial need in Mexico: the automated classification of chili peppers according to official quality norms. This focus on solving real problems with available tools, rather than chasing benchmark performance, represents a fundamental characteristic of Latin American research philosophy.

Similarly, the work of [38] on pineapple ripening assessment in Colombia indicates how sophisticated results can be achieved through clever application of relatively simple techniques. By employing HSV color space analysis and Otsu thresholding, they achieved high classification accuracy for Perolera pineapples. This study exemplifies how Latin American researchers have developed deep expertise in classical computer vision tech-

niques, achieving impressive results without the computational resources required for training large neural networks.

5.2.2. Building Complete Systems from Accessible Components

One of the most striking features of Latin American research is the emphasis on creating complete, integrated systems using widely available, low-cost hardware. This approach is particularly well exemplified by [37], who developed a comprehensive quality management system for Cavendish bananas using Raspberry Pi and Arduino components. Their work involved capturing over a thousand augmented images over six months, processing them with a combination of classical techniques and machine learning regression models. The resulting mobile application could predict remaining shelf life in real conditions, addressing a genuine need in Colombia's banana export industry.

This systems-level thinking is further demonstrated by [35] in their work on passion fruit mass and volume estimation. By combining multiple feature extraction methods with PCA and LDA, they created a comprehensive system that could estimate both mass and volume from digital images. The integration of these various techniques into a coherent workflow, implemented using open-source tools like OpenCV and QT Creator, reflects the Latin American emphasis on creating practical, deployable solutions rather than isolated technological demonstrations.

5.2.3. The Agricultural Context: A Focus on Export Crops

The crops selected for study in Latin America reveal much about the region's agricultural priorities and economic dependencies. The focus on export-oriented fruits such as bananas [37], pineapples [38], passion fruit [35], and chili peppers [42] reflects the importance of agricultural exports to Latin American economies. This focus has both advantages and limitations. On one hand, it ensures that research is directly relevant to major economic sectors and addresses genuine industrial needs. On the other hand, it may lead to a neglect of subsistence crops and smallholder agriculture, which remain critically important for food security and rural livelihoods in many Latin American countries.

The research on dried white mulberry by [63] represents an interesting variation on this pattern. While conducted in Iran rather than Latin America, the approach shares many characteristics with Latin American research: the use of accessible RGB cameras, manual expert grading as a gold standard, and the successful application of ANNs and SVMs to achieve perfect classification accuracy. This work indicates that the pragmatic, resource-conscious approach characteristic of Latin American research has global relevance for agricultural contexts where advanced sensing infrastructure is not available.

5.3. *The International Approach: Pushing Technological Boundaries*

The international corpus analyzed in this review should not be regarded as a homogeneous body of research. Rather, it encompasses studies conducted in countries with diverse agricultural practices, economic resources, technological infrastructures, and research priorities. Consequently, the patterns identified in this section should be interpreted as trends emerging from the reviewed literature rather than as universal characteristics of all international research. This distinction is particularly important because many small- and medium-scale producers outside Latin America also face challenges related to affordability, accessibility, local adaptation, and limited technological resources, despite operating in different geographical contexts.

In contrast to the pragmatic resource-conscious philosophy that characterizes Latin American research, the international research community—particularly in technologically advanced nations and rapidly developing economies like China—has embraced a paradigm defined by technological ambition and boundary-pushing innovation. This approach is

distinguished by the adoption of state-of-the-art deep learning architectures, the integration of sophisticated sensing modalities such as hyperspectral imaging and 3D scanning, and the development of complete robotic systems that automate the entire agricultural pipeline from field detection to harvesting and sorting. Fueled by greater access to computational resources, larger research budgets, and stronger connections to commercial technology sectors, this research culture has produced remarkable advances in accuracy, efficiency, and system integration. However, this pursuit of technological excellence also raises important questions about accessibility, practicality, and the potential disconnect between cutting-edge research and the needs of the majority of the world's agricultural producers.

5.3.1. The Pursuit of Cutting-Edge Performance

Research conducted outside Latin America, particularly in technologically advanced nations and rapidly developing economies like China, exhibits a markedly different character. Here, the emphasis is squarely on pushing the boundaries of what is technologically possible, employing the most sophisticated algorithms and sensing modalities available. This research culture is supported by greater access to computational resources, larger research budgets, and stronger connections to commercial technology sectors.

The work on vision transformers (ViT) by [58,60] exemplifies this pursuit of cutting-edge performance. Rather than relying on classical computer vision techniques, these researchers have adopted the latest deep learning architectures, originally developed for general computer vision tasks, and adapted them for fruit quality assessment. The Multi-Task NoisyViT developed by Esfandiari Fard and colleagues, which achieves near-perfect accuracy for both fruit type classification and freshness detection, represents the kind of sophisticated multi-task learning that is only possible with substantial computational resources and large datasets.

Similarly, the EMBS-DETR architecture developed by Yan et al. [52] for tomato maturity detection indicates how researchers outside Latin America are able to combine multiple advanced techniques—frequency-domain convolution, parameter-free attention mechanisms, and multi-scale feature fusion—to achieve exceptional performance while simultaneously addressing computational efficiency. This simultaneous pursuit of both accuracy and efficiency reflects a maturity in the research ecosystem that allows for optimization across multiple dimensions.

5.3.2. The Integration of Robotics and Automation

Perhaps the most dramatic distinction between international and Latin American research lies in the integration of fruit detection technologies with robotics and automation systems. While Latin American research has largely focused on quality assessment as a standalone task, international research increasingly addresses the complete agricultural automation pipeline from field detection to robotic harvesting and sorting.

The lithic picking robot developed by Wang et al. [50] represents this integrated approach in its most sophisticated form. This system combines YOLOv8-based fruit and branch segmentation with binocular stereo vision for 3D localization, D-H inverse kinematics for robotic arm control, and an innovative air-blowing obstruction removal unit. The result is a complete robotic system capable of navigating natural environments, identifying obstructions, and successfully harvesting fruit clusters. This indicates that such integrated systems are approaching practical viability.

The robotic apple sorting system developed by Zhang et al. [55] similarly indicates the power of integrating sensing with physical automation. Their pitch-variable screw conveyor system, combined with machine vision for size and color grading, achieved remarkable sorting accuracy exceeding 99% even at high throughputs. This level of performance,

validated with nearly a thousand apples across multiple varieties and operating modes, represents the kind of industrial-ready system that is the ultimate goal of agricultural automation research.

5.3.3. Sensor Fusion: Combining Multiple Modalities

International research has also led the way in combining multiple sensing modalities to achieve more comprehensive fruit assessment. While Latin American research has primarily relied on conventional RGB imaging, international studies increasingly incorporate near-infrared spectroscopy, hyperspectral imaging, 3D scanning, and other advanced sensing techniques.

The work of Zhao et al. [56] on apple internal quality assessment exemplifies this multi-modal approach. By combining SSD-based machine vision for target detection with Vis/NIR spectroscopy for internal quality measurement, they created a robotic system capable of simultaneously grasping fruit, detecting spectral characteristics, and classifying by soluble solids content. The integration of these complementary sensing modalities allowed for comprehensive quality assessment, including both external appearance and internal properties.

Similarly, Hu et al. [64] demonstrated the power of 3D sensing for bruise detection in apples. By converting irregular 3D mesh data from NIR scanning into 2D feature maps, they were able to apply state-of-the-art CNNs to the classification problem, achieving exceptional accuracy. This innovative approach to data representation allowed them to leverage the full power of deep learning for a fundamentally 3D sensing problem, highlighting how advanced sensing technologies enable new analytical approaches.

Future research should move beyond conventional RGB image analysis toward multimodal sensing systems integrating computer vision with spectroscopy, biosensing, and other physics-based sensing technologies. Such sensor fusion approaches could substantially improve robustness by simultaneously evaluating external appearance, internal physiological conditions, and biochemical composition. Artificial intelligence will play a central role in integrating heterogeneous data streams into unified decision-support systems suitable for precision agriculture and automated quality inspection.

5.3.4. Edge Computing and Deployment Readiness

A notable trend in international research is the growing emphasis on making sophisticated AI systems deployable on edge computing devices. This focus on practical deployment, while still requiring substantial technical sophistication, represents an important convergence with the pragmatic concerns that have always motivated Latin American research.

The work of Shi et al. [45] on blueberry maturity detection exemplifies this trend. By applying random channel pruning and TensorRT FP16 quantization, they reduced the model size and achieved real-time inference on an NVIDIA Jetson Orin Nano. This remarkable efficiency allows the system to operate in natural orchard environments with limited computational resources, bringing sophisticated AI capabilities to the field.

Similarly, the TinyML-based shelf life estimator developed by [69] indicates how extreme efficiency can be achieved. By deploying a neural network regression model on an Arduino Nano 33 BLE Sense microcontroller, they created a handheld device that can predict date fruit freshness with remarkable accuracy using minimal memory and power consumption. This approach represents a vision of truly democratized agricultural AI that could have enormous impact in both developed and developing economies.

While cloud-based AI platforms provide virtually unlimited computational resources, they frequently depend on stable Internet connectivity, centralized data transmission, and remote processing. These requirements are often difficult to satisfy in agricultural

environments, particularly in rural regions where network infrastructure may be limited or intermittent.

Edge computing addresses these limitations by executing AI inference directly on embedded devices located near the sensing platform, such as drones, mobile robots, autonomous tractors, smartphones, Raspberry Pi systems, NVIDIA Jetson modules, or TinyML microcontrollers. By eliminating continuous communication with remote servers, edge computing substantially reduces inference latency, communication costs, and bandwidth requirements while improving data privacy and operational reliability.

Another important advantage of edge computing is its ability to provide real-time decision support. Fruit harvesting, robotic manipulation, grading, and quality inspection frequently require immediate responses that cannot tolerate delays associated with cloud communication. Local inference therefore enables rapid detection of fruit maturity, defects, and harvesting readiness while simultaneously reducing dependence on Internet availability.

From a sustainability perspective, edge computing also contributes to lower energy consumption by reducing continuous data transmission between sensing devices and cloud servers. Furthermore, distributed processing improves system resilience because agricultural operations can continue even during temporary communication failures. Nevertheless, cloud computing remains valuable for computationally intensive tasks such as large-scale model training, centralized data management, and long-term agricultural analytics. Consequently, future intelligent agricultural systems will likely adopt hybrid cloud–edge architectures that combine the strengths of both paradigms.

5.4. Critical Analysis: Strengths, Weaknesses, and Blind Spots

Having examined the distinct technological philosophies and research priorities that characterize Latin American and international approaches to fruit detection, it is essential to subject both traditions to critical scrutiny. Each approach possesses genuine strengths that deserve recognition—the Latin American model’s commitment to accessibility and contextual relevance, and the international model’s technological sophistication and systems integration. Yet, both traditions also face significant limitations that constrain their impact and reach. More troubling still, both share critical blind spots that have been systematically neglected across the literature: the near-total absence of attention to human factors such as AI-assisted decision support, the limited consideration of inclusivity and accessibility for diverse user populations, and the persistent failure to address the environmental sustainability of AI-based agricultural systems. This critical analysis seeks to illuminate these strengths, weaknesses, and omissions, providing a foundation for charting a more balanced and responsible path forward.

5.4.1. The Latin American Model: Strengths and Limitations

The Latin American approach to fruit detection research possesses distinctive strengths that deserve recognition. First and foremost, the focus on practical, deployable solutions using accessible technology ensures that research outcomes can actually benefit agricultural practitioners. The systems developed by researchers like [35,37] are not merely academic exercises but represent genuine attempts to address industry needs within realistic technological and economic constraints.

Second, the depth of expertise in classical computer vision techniques that Latin American researchers have developed is itself a valuable resource. These foundational techniques remain relevant, particularly for applications where large training datasets or substantial computational resources are not available. The ability to achieve high accuracy with relatively simple approaches, as demonstrated by [38,39], represents a form of technological wisdom that should not be undervalued.

Third, the emphasis on crops of regional economic importance ensures that research directly benefits local agricultural sectors. This contextual relevance is often lacking in international research, which may focus on crops or conditions that are not representative of agricultural realities in developing economies.

However, the Latin American approach also faces significant limitations. The relative scarcity of advanced sensing equipment—hyperspectral cameras, NIR spectrometers, 3D scanners—limits the range of problems that can be addressed. While remarkable results can be achieved with conventional RGB imaging, certain quality attributes can only be assessed with more sophisticated sensing modalities.

Additionally, the smaller datasets and simpler models used in much Latin American research may limit generalizability. Systems developed with hundreds or thousands of samples may not perform as robustly when deployed in diverse conditions as systems trained on tens of thousands of images from multiple environments. The challenge of achieving reliable performance across the full range of natural variability remains a significant hurdle.

5.4.2. The International Model: Achievements and Excesses

The international approach to fruit detection research has achieved remarkable technological feats that should not be dismissed as mere academic exercises. The development of architectures like [52] EMBS-DETR and [45] BR-DETR-Prune, achieving exceptional accuracy with efficient deployment, represents genuine engineering achievement. These systems are approaching the point where practical deployment in commercial agriculture becomes feasible.

The integration of robotics with sensing technologies, as demonstrated by [50,55], represents perhaps the most significant advance in agricultural automation. These systems demonstrate that complete automation of harvesting, sorting, and quality assessment is technically achievable, with profound implications for agricultural labor, food waste reduction, and supply chain efficiency.

However, the international approach is not without its excesses. The pursuit of incremental improvements in classification accuracy, often at substantial computational cost, raises questions about the practical value of such efforts. When a system already achieves high accuracy, is the additional research effort required to reach even higher performance genuinely worthwhile? In many cases, the constraints on practical deployment are not technological but economic, social, and institutional. Moreover, the high accuracy values reported throughout this review should not be interpreted as the performance of a single unified detection model. Instead, they represent outcomes reported independently by the original studies under their respective experimental conditions. Therefore, the objective of this review is to critically compare published evidence rather than to estimate the predictive performance of a novel AI algorithm.

There is also a risk that the international research agenda becomes disconnected from the needs of the majority of the world's agricultural producers. The focus on high-value crops, advanced sensing equipment, and sophisticated automation may produce solutions that are inaccessible to smallholders and farmers in developing economies. This disconnect, if not addressed, threatens to exacerbate agricultural inequalities rather than reduce them.

5.4.3. Research Funding as a Driver of Technological Development

One factor that may partially explain the technological differences observed between Latin American and international studies is the availability of research funding. Although a formal quantitative comparison of funding sources was beyond the scope of this review, qualitative examination of the included studies suggests that many international projects benefit from large-scale governmental programs, national innovation

agencies, and industry-supported initiatives. Such financial support facilitates access to advanced sensing equipment, robotic platforms, high-performance computing infrastructure, and multidisciplinary research teams.

Conversely, several Latin American studies emphasize the development of affordable solutions based on commercially available hardware, including Raspberry Pi, Arduino, conventional RGB cameras, and open-source software frameworks. Rather than representing a technological weakness, this strategy reflects an adaptation to local economic conditions and indicates a strong commitment to developing deployable solutions suitable for regional agricultural contexts.

Future systematic reviews could explicitly extract funding information and quantitatively investigate possible relationships between funding sources, technological sophistication, publication impact, and deployment readiness across different geographical regions.

5.4.4. Shared Blind Spots

Remarkably, both Latin American and international research share some significant blind spots. The near-total neglect of AI-assisted decision support in the technical literature, even as a human factors consideration, represents a striking omission. Given that the ultimate goal of agricultural automation is to augment human capabilities, attention to how such systems affect and interact with human competencies—including mathematical and analytical skills—seems essential.

Similarly, the limited attention to inclusivity and accessibility in technical development is troubling. The observation that only two studies address learning difficulties among students or agricultural professionals suggests that the human dimensions of technology adoption are being overlooked. If automated fruit detection systems are to be adopted by diverse populations with varying educational backgrounds and abilities, their design must account for human factors beyond simple usability.

The environmental sustainability of AI-based agricultural systems is another area of neglect. While many studies demonstrate impressive performance, few consider the energy consumption of training and deploying AI models, the lifecycle environmental impacts of electronic components, or the potential for technology to enable more sustainable agricultural practices. As the environmental costs of AI become increasingly recognized, this blind spot will demand urgent attention.

5.5. *Why High Laboratory Accuracy Does Not Always Translate to Field Performance*

Although many reviewed studies reported classification accuracies exceeding 95%, these values were frequently obtained using carefully curated datasets collected under controlled laboratory conditions. Such datasets often exhibit homogeneous illumination, standardized camera positioning, limited background complexity, and balanced class distributions. Consequently, trained models learn highly discriminative features that may not remain robust under real agricultural conditions.

Field deployment introduces substantially greater variability, including changing sunlight intensity, shadows, rain, dust, wind-induced fruit motion, partial occlusions caused by leaves or branches, variable viewing angles, differences in fruit maturity, seasonal changes, sensor noise, and camera calibration inconsistencies. These factors collectively generate distribution shifts between training and operational environments, causing degradation of model performance.

Another important challenge concerns dataset representativeness. Many publicly available fruit datasets contain thousands of carefully labelled images captured under ideal acquisition conditions but fail to represent the full diversity encountered in commercial

orchards. Consequently, models frequently exhibit overfitting to laboratory environments despite excellent validation metrics.

Robust deployment therefore requires not only larger datasets but also more representative datasets collected under diverse environmental conditions. Data augmentation, domain adaptation, transfer learning, self-supervised learning, continual learning, and multimodal sensor fusion have emerged as promising strategies for improving generalization across heterogeneous agricultural scenarios.

Future research should prioritize long-term field validation involving multiple orchards, seasons, climatic conditions, fruit varieties, and imaging devices. Benchmark datasets should progressively evolve from laboratory collections toward realistic agricultural datasets capable of better representing operational deployment conditions.

5.6. The Path Forward: Integration, Democratization, and Human-Centered Design

The critical analysis of Latin American and international research traditions suggests not a hierarchy of superior and inferior approaches, but rather complementary strengths that, if thoughtfully integrated, could transform the field of fruit detection technology. The path forward lies not in one paradigm supplanting the other, but in a deliberate synthesis that harnesses international technical sophistication while preserving Latin American commitments to accessibility and contextual relevance. This vision of integration rests on three interconnected pillars: the democratization of advanced technologies through knowledge transfer and edge computing, the continued cultivation of locally adapted solutions that address regional agricultural needs, and a fundamental reorientation toward human-centered design that prioritizes educational outcomes, inclusivity, and environmental sustainability. By pursuing this integrated approach, researchers can develop technologies that are not merely more accurate or efficient, but genuinely transformative for agricultural communities worldwide.

5.6.1. Technology Democratization and Knowledge Transfer

The most promising future direction lies in democratizing the advanced technologies developed in international research and making them accessible to the Latin American research community and agricultural sector. The recent trend toward edge computing and model efficiency, exemplified by [45,69], is particularly encouraging in this regard. These developments show that sophisticated AI can be deployed on increasingly affordable hardware, making advanced sensing and analysis capabilities accessible to resource-constrained users.

Knowledge distillation, as demonstrated by [44] in their work using YOLOv8, offers a pathway for transferring capabilities from large models to lightweight implementations. By using a larger teacher model to guide a smaller student model, researchers can achieve performance approaching that of state-of-the-art systems with dramatically reduced computational requirements. This approach could allow Latin American researchers to leverage the developments of the international research community while operating within their own resource constraints.

5.6.2. Contextual Adaptation and Indigenous Innovation

However, technology transfer is not sufficient. The Latin American research community must continue to develop locally adapted solutions that address the specific needs of the region's agricultural sectors. The focus on regionally important crops, the use of accessible hardware, and the integration of systems for real agricultural contexts are strengths that should be preserved and enhanced.

There is particular opportunity in developing open-source platforms and shared datasets that can support research across the region. The success of open-source tools

like OpenCV and Python libraries in Latin American research indicates the power of collaborative, accessible resources. Building on this foundation, regional research networks could create shared repositories for agricultural imagery, develop standardized benchmarks, and coordinate research efforts to maximize impact.

5.6.3. Human-Centered AI, Inclusivity, and Sustainability

Although considerable progress has been achieved in improving the technical performance of AI-enabled fruit detection systems, future agricultural innovation should not be evaluated solely according to prediction accuracy. Intelligent sensing technologies must also address broader societal, economic, and environmental challenges to ensure long-term adoption and sustainable impact.

Human-centered AI represents an important research direction because agricultural practitioners require more than numerical predictions. Farmers, inspectors, and supply chain operators benefit from interpretable decision-support systems capable of explaining why fruits are classified into particular quality categories. Such explainability improves user confidence, facilitates technology adoption, and supports transparent agricultural decision-making.

Inclusivity is equally important because many agricultural regions, particularly in developing countries, have limited access to high-performance computing infrastructure, expensive sensing equipment, and specialized technical expertise. Consequently, future intelligent agricultural systems should be designed to operate on affordable hardware platforms, support multiple languages, accommodate users with different levels of digital literacy, and remain accessible to smallholder farmers as well as large agricultural enterprises.

Environmental sustainability also represents a fundamental consideration for next-generation AI-enabled agriculture. Intelligent sensing systems can contribute to reducing food waste through earlier detection of defects and optimal harvesting decisions while simultaneously decreasing unnecessary pesticide application, improving resource utilization, reducing transportation losses, and supporting precision agriculture practices. Furthermore, energy-efficient AI models and edge computing architectures contribute to reducing the environmental footprint associated with large-scale artificial intelligence deployment.

Future research should therefore evaluate intelligent agricultural systems using multidimensional criteria that extend beyond predictive accuracy, including interpretability, accessibility, affordability, environmental impact, deployment readiness, and long-term societal benefits.

6. Limitations of This Scoping Review

Several limitations should be considered when interpreting the findings of this scoping review.

- First, the evidence base was characterized by substantial methodological heterogeneity. The included studies differed considerably in sensing technologies, computational approaches, target fruit species, dataset characteristics, validation protocols, evaluation metrics, and experimental settings. This diversity precluded direct quantitative comparison across studies and made a formal meta-analysis inappropriate. Consequently, the present review provides a qualitative synthesis of the available evidence rather than a statistical comparison of algorithmic performance.
- Second, most included studies evaluated their proposed methods using curated laboratory datasets or controlled experimental environments. Although these studies frequently reported high predictive performance, such results may not fully reflect the variability encountered under real agricultural conditions, including changing illumination, occlusion, weather conditions, fruit maturity variability, and differences in

acquisition hardware. Therefore, reported performance metrics should be interpreted with appropriate caution.

- Third, the geographical distribution of the included studies was uneven, with only eight studies originating from Latin America. Accordingly, the regional comparisons presented in this review should be interpreted as patterns emerging from the available evidence rather than as definitive representations of all research conducted in Latin America or internationally.
- Fourth, although a comprehensive search strategy was implemented across Scopus, IEEE Xplore, and Google Scholar, the exceptionally large number of retrieved records (242,310) required the adoption of a pragmatic screening strategy in which only the first 300 relevance-ranked records from each database were screened. While this approach has been adopted in previous scoping reviews facing similarly large retrieval volumes, potentially relevant studies appearing beyond this threshold may have been missed. Consequently, the present review should be interpreted as a synthesis of the most highly ranked and accessible evidence rather than as an exhaustive representation of the entire body of published literature.
- Finally, although this review identifies emerging themes related to AI-assisted decision support, inclusivity, environmental sustainability, and human-centered system design, these aspects were explicitly addressed in only a limited number of the included studies. Therefore, these observations should be interpreted as emerging research opportunities rather than definitive conclusions regarding the current state of the field.

Despite these limitations, the authors believe that the present scoping review makes a meaningful contribution to understanding the technological landscape of fruit detection and quality assessment, the regional disparities in research capacity and priorities, and the critical gaps that must be addressed to achieve a more equitable, sustainable, and impactful agricultural technology future. These limitations should not be viewed as fatal flaws but rather as inherent characteristics of scoping review methodology and as opportunities for future research to address the identified gaps with more rigorous, comprehensive, and quantitative approaches.

7. Conclusions and Future Work

7.1. Conclusions

This scoping review critically examined 35 studies on sensor technologies and artificial intelligence for fruit detection, classification, and quality assessment, synthesizing evidence from both Latin American and international research traditions. The analysis suggests a rich and diverse technological landscape, yet one characterized by significant disparities in resources, priorities, and research cultures that have profound implications for agricultural development worldwide. Importantly, the regional differences identified in this review should not be interpreted as rigid distinctions between Latin America and the rest of the world. Similar technological, economic, and infrastructural challenges are also encountered by many smallholder and medium-scale producers across Europe, Asia, Africa, and other regions. Consequently, affordable, adaptable, and human-centered AI solutions represent a global priority rather than a challenge unique to any particular geographical area.

The contrast between Latin American and international approaches to fruit detection technology is not a competition between inferior and superior methods, but a reflection of different contexts, priorities, and constraints. Latin American research tends to emphasize developing accessible, practical, and contextually relevant solutions, demonstrating remarkable ingenuity with limited resources. International research pushes the boundaries of what is technologically possible, creating the sophisticated algorithms and integrated systems that represent the future of agricultural automation. The greatest opportunities lie

not in one approach dominating the other, but in their integration. The democratization of international research advances, the contextual adaptation of technologies for local needs, and the mutual recognition of strengths and limitations can create a virtuous cycle of innovation. By combining international technical sophistication with Latin American pragmatic problem-solving, we can develop fruit detection technologies that are both powerful and accessible, sophisticated and sustainable, advanced and human-centered.

The ultimate goal is not simply better classification accuracy or more efficient automation, but the transformation of agricultural practices in ways that enhance food security, economic development, and environmental sustainability. Achieving this goal will require the full participation of researchers from both Latin America and the international community, working together to ensure that the benefits of sensor technology and AI are shared equitably across the global agricultural landscape. Future AI-enabled agricultural platforms are expected to increasingly adopt hybrid edge–cloud architectures capable of combining real-time local inference with centralized model training and long-term data analytics.

7.2. Future Work

Several limitations of this scoping review should be acknowledged. First, although the multi-database search strategy and relevance-based screening were designed to balance comprehensiveness with feasibility, the screening of only the first 300 relevance-ranked records from each database may have excluded relevant studies. While previous methodological work supports this approach for mature research topics, the potential for missing less-cited but relevant studies cannot be excluded. Second, the present review did not perform a formal meta-analysis, and the reported performance metrics should therefore be interpreted as indicative rather than definitive comparative evidence. Third, research funding information was not systematically extracted during protocol development, precluding a statistical comparison of financial support mechanisms between Latin American and international research. Future reviews could address these limitations by employing more comprehensive screening strategies and including formal funding analyses.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/computers15080518/s1>, Table S1: PRISMA-2020-checklist.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
ANN	Artificial Neural Network
API	Application Programming Interface
AR	Augmented Reality
BiFPN	Bidirectional Feature Pyramid Network
CCD	Charge-Coupled Device
CNN	Convolutional Neural Network
DOI	Digital Object Identifier
ELU	Exponential Linear Unit
F1	F1-Score
FoV	Field of View
FPS	Frames Per Second
GAP	Global Average Pooling
GLCM	Gray-Level Co-occurrence Matrix
GPR	Gaussian Process Regression
GPU	Graphics Processing Unit
HSV	Hue, Saturation, Value
IoT	Internet of Things
LBP	Local Binary Pattern
LDA	Linear Discriminant Analysis
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
mAP	Mean Average Precision
MIS	Multispectral Imaging
MLP	Multilayer Perceptron
MMAT	Mixed Methods Appraisal Tool
MSE	Mean Squared Error
NIR	Near-Infrared
OSF	Open Science Framework
PCA	Principal Component Analysis
PLS	Partial Least Squares
RBF	Radial Basis Function
RGB	Red, Green, Blue
RMSE	Root Mean Square Error
ROI	Region of Interest
SSC	Soluble Solids Content
SSD	Single Shot Multibox Detector
SVM	Support Vector Machine
ViT	Vision Transformer
VR	Virtual Reality
XAI	Explainable Artificial Intelligence
YOLO	You Only Look Once

Appendix A. The Selected Studies

The following section presents the results of this scoping review, synthesized from the 35 studies (8 from Latin America and 27 outside Latin America) that met our inclusion criteria. A comprehensive overview of these studies, including their main goals, are presented in Tables A1 and A2. These tables serve as the foundation for the subsequent thematic analysis organized around the research questions.

Table A1. Studies related to Latin American initiatives for fruit classification systems included in the scoping review.

Authors	Main Goal	Population	Computational Techniques	Gold Standard	Outstanding Accuracy
Cruz-Domínguez et al. [42]	Develop an ANN-based classification system for dried chili pepper size and color recognition according to Mexican Official Norm NMX-FF-107/1-SCFI-2014	850 Guajillo chili samples (extra, first, and second class)	8-bit grayscale histograms, MLP neural network with 256 input neurons, 2 hidden neurons, 1 output neuron, backpropagation (BFGS quasi-Newton), MATLAB	Mexican Official Norm NMX-FF-107/1-SCFI-2014 (expert farmers selected 850 samples as reference standard)	Overall accuracy 82.13% (AUC), 82.7% (confusion matrix), Cohen's kappa 0.669
Constante et al. [41]	Develop an artificial vision system with deep neural architecture and noise-compensated learning for strawberry classification by shape, color, and anomalies for industrial use	50 images (40 high-quality strawberries for training, 10 undesirable/foreign objects) + real-time testing	Canny edge detection, moving ROI (100 × 100), deep neural architecture (two-stage: tracking agent + 12 ANEAs with 10,000 inputs, 37 hidden, 5 output neurons, sigmoidal, backpropagation), noise-compensated learning (white noise as counterexample), OpenCV, MATLAB	Human perceptual evaluation (40 volunteers selected visually appealing strawberries from 100 images as reference standard)	Extra: 92.5% (37/40); Consumption: 90% (36/40); Raw material: 90% (36/40); Foreign objects: 100% (40/40)
Alvarado and Molina [40]	Train a deep convolutional neural network (MobileNetV1) for fruit classification across 13 categories using TensorFlow	13,516 images across 13 fruit categories: banana, cherry, strawberry, lemon, mandarin, mango, apple, blackberry, orange, papaya, pear, pineapple, and grape	MobileNetV1 (CNN), TensorFlow, Python, bottleneck feature extraction, 4000 training steps, image resizing (500 × 500 to 3120 × 4160 pixels), JPG/JPEG format	N/A (no explicit baseline comparator; model trained from scratch with transfer learning)	87% overall classification accuracy; average recognition speed 0.406–0.463 s per image; best performance for banana, cherry, strawberry, mandarin, blackberry, papaya, pineapple, and grape; regular performance for mango, apple, orange, and pear; poor performance for lemon
Silva and Lizcano [38]	Develop an automatic computer vision methodology to evaluate the post-harvest ripening stage of pineapple (Ananas Comosus) Perolera variety	319 images of Perolera pineapples (110 for training, 209 for evaluation); 1500 total images captured over 11-day ripening period	HSV color space, Otsu automatic thresholding segmentation, first-order moment (mean) of H and S channel histograms as features, MBSAS clustering algorithm (unsupervised classification), median filter (moving average, window = 9), OpenCV library	Three expert evaluators with >10 years of experience in pineapple cultivation and commercialization (manual classification consensus)	96.36% overall classification accuracy; Stage 1 (green): 100%; Stage 2 (pintona): 100%; Stage 3 (ripe): 88.24% (2 misclassifications out of 17); average recognition speed suitable for online applications
Gonzalez et al. [35]	Develop a computer vision system to estimate mass and volume of passion fruit (Passiflora edulis) using digital image features	100 passion fruit images (80 training, 20 validation); mass range 25.00–118.50 g, volume range 99.37–231.64 cm ³	HSV and CIELab color spaces, OHTA segmentation ($I_2 = (R - B) / 2$), GLCM texture features (Entropy, Homogeneity, Contrast, Energy, Correlation), Hu invariant moments (shape), PCA (8 components, 96.5% variability), LDA (6 components, 96.5% discriminance), RLM, ANN (8 inputs, 6 hidden neurons, 1 output, 2/3 rule, sigmoidal, cross-validation), OpenCV, Qt Creator	Traditional laboratory measurements (digital balance for mass, ellipsoid model $V = 4\pi/3 \cdot abc$ for volume as reference standard)	Mass estimation: $R = 0.83$ (test), $R = 0.77$ (training), typical error 15.97 g (test), 16.59 g (training); Volume estimation: $R = 0.73$ (test), $R = 0.75$ (training), typical error 31.58 cm ³ (test), 34.44 cm ³ (training); Classification accuracy (low/medium/high classes): mass 86.6% (test), volume 80.0% (test); LDA+ANN outperformed PCA+RLM by >20%
Larcher et al. [36]	Develop a computer vision system to detect and quantify external skin damage (defects/bruises) in pears for quality grading and postharvest inspection	176 images (4 faces of 22 pears, 2 undamaged control fruits, 20 with varying damage degrees; two sessions over one week to accentuate deterioration)	RGB color space, grayscale conversion, global thresholding, background removal using R and B channel thresholds ($R > 150$, $B > 100$), connected-component labeling algorithm (recursive), lenticle filtering (size-based threshold to exclude natural spots), area calculation, damage percentage estimation, MATLAB with Image Processing Toolbox	Human visual inspection (traditional manual quality assessment by trained inspectors as reference standard)	Damage percentage estimation: 22.54%, 20.74%, 26.32%, 36.88% (quantified defect areas shown in results); system successfully detects and measures external defects; limitations include distinguishing stem/calyx from true defects and handling shadows/glare from directional lighting
Nieto and Rangel [37]	Develop a computer vision system for post-harvest quality management of Cavendish banana, estimating ripeness stage and remaining shelf life (days until spoilage) using machine learning	1093 augmented images of Cavendish bananas (multiple ripening rounds over 6 months); captured using IoT system (Raspberry Pi 3, Arduino Nano, Camera Module V2, DHT22 sensor)	RGB histogram percentage descriptor, Canny edge detection, Gaussian filtering, ROI extraction, image augmentation (rotations 15–75°), mosaics (4 × 4 tiles, 600 × 600 px), binary detection model (LBP + KNN, 99.37% accuracy), regression models: SVR, Random Forest Regressor, MLP Regressor; Python (OpenCV, Scikit-learn, NumPy), Flask API, React Native mobile app	Expert visual inspection (manual ripeness estimation by experienced merchants and consumers as reference standard); Colombian technical standards NTC5139 (mango) and NTC4101 (granadilla) referenced as analogous baselines	SVR: MSE = 13.43, MAE = 2.74, RMSE = 3.66; RFR: MSE = 3.86, MAE = 1.49, RMSE = 1.97; MLPR: MSE = 3.80, MAE = 1.52, RMSE = 1.95; Binary detection: 99.37% accuracy, 0.974 precision, 1.0 recall; Real-world validation: 3/4 bananas reached spoilage within predicted days; mobile app successfully predicts remaining shelf life in real conditions

Table A1. Cont.

Authors	Main Goal	Population	Computational Techniques	Gold Standard	Outstanding Accuracy
Sucari et al. [39]	Determine the effectiveness of applying artificial vision in pattern recognition for fruit classification in agribusiness using a Bayesian classifier	50 database records of 6 fruit varieties (avocado, tuna/prickly pear, pear, mango, orange, granadilla) with 4 characteristics each (size, shape, color, texture); 20 fruit samples per variety for testing (2 lots of 20 each)	RGB and HSV color spaces, image segmentation via thresholding on blue channel (RGB), feature extraction (area, perimeter, roundness, major axis, minor axis, mean values of HSV and RGB channels), Bayesian classifier (multivariate normal density function), implemented in GNU Octave	Traditional manual classification (human inspection as reference standard); KNN classifier (reported in literature) as baseline comparator (Bayesian outperformed KNN)	Overall classification accuracy: 93.33%; Orange: 97.5%; Granadilla: 97.5%; Tuna (prickly pear): 95.0%; Avocado: 92.5%; Pear: 87.5%; Error rate: 6.67%.

Table A2. Studies related to Non-Latin American initiatives for fruit classification systems included in the scoping review.

Authors	Main Goal	Population	Computational Techniques	Gold Standard	Outstanding Accuracy
Wanthong et al. [51]	Develop a low-cost, real-time system using a single 2D camera and AI to estimate 3D dimensions (width, height, depth) of 'Nam Dok Mai Si Thong' mangoes for optimizing X-ray irradiation dose, reducing energy consumption and operational costs.	84 export-grade 'Nam Dok Mai Si Thong' mangoes (Size classes 1 L, 2 L, 3 L) from three sources in Thailand.	Computer Vision (Canny edge detection), Linear Regression (LR), Artificial Neural Network (ANN), ANN with Optuna hyperparameter optimization, Gaussian Process Regression (GPR), and Co-Kriging (LR+CoK) for multi-fidelity depth estimation.	ANN (Default) with depth from LR: 86.67%.	ANN (Default) with depth from LR+CoK: 93.33%; ANN+Optuna with depth from LR+CoK: 93.33%; GPR with depth from LR+CoK: 93.33%.
Apostolopoulos et al. [58]	Develop a general machine learning model based on Vision Transformers (ViT) for assessing fruit quality (good vs. rotten) across multiple fruit types using visual appearance alone, eliminating the need for fruit-specific models.	16 fruit types (Apple, Banana, Cucumber, Grape, Guava, Kaki, Lemon, Lime, Mango, Orange, Papaya, Peach, Pear, Pomegranate, Tomato, Watermelon) with a total of 29,215 images in the Union Dataset (UD).	Vision Transformer (ViT) with multi-head self-attention, patch encoding, and MLP layers; 10-fold cross-validation; comparison with 25+ state-of-the-art CNNs and classical ML models (Random Forest, XGBoost, SVM, KNN, Neural Network).	Classical ML (Random Forest): 93.43%; State-of-the-art CNN (ResNet152): 97.85%.	ViT General Model (10-fold CV on UD): 97.94%; External per-fruit evaluation: up to 100% (Grapes), 99.50% (Apples, Kakis, Oranges, Tomatoes), 99% (Cucumbers), 98% (Papayas, Peaches, Watermelons).
Yan et al. [52]	Develop a lightweight end-to-end detection model (EMBS-DETR) for tomato maturity detection in complex field environments with illumination variation, severe occlusion, and large fruit scale differences, achieving high accuracy with low computational cost for edge deployment.	2327 real-field images of pink large-fruit tomatoes (immature: 796, turning: 718, mature: 813) collected from two planting bases in Shanxi Province, China; expanded to 3289 images via augmentation.	EMBS-DETR (based on RT-DETR-R18) with C2f-FDConv backbone (frequency-domain convolution), SimAM-TF parameter-free attention mechanism, and EMBS-FPN multi-scale feature fusion with BiFPN weighted fusion and multi-branch receptive field modeling.	RT-DETR (baseline): 87.4% Precision, 84.3% Recall, 87.2% mAP@0.5, 78.4% mAP@0.5:0.95, 75.82 MB parameters, 56.9 GFLOPs.	EMBS-DETR: 90.9% Precision, 85.7% Recall, 89.9% mAP@0.5, 79.8% mAP@0.5:0.95, with only 37.03 MB parameters (−51.1%) and 25.2 GFLOPs (−55.7%).
Wang et al. [50]	Develop a litchi picking robot system with active obstruction removal capability using AI-based visual perception to locate picking points, identify obstructions, and guide the end-effector to successfully harvest litchi clusters in natural environments.	1100 litchi images (712 outdoor, 288 indoor) from Zengcheng District, Guangzhou, China; 16 litchi clusters for obstruction removal experiments.	YOLOv8-Seg for fruit and branch segmentation; binocular stereo vision for 3D localization; 24-pixel field feature analysis for obstruction identification; D-H inverse kinematics for robotic arm control; air-blowing obstruction removal unit.	YOLOv8-Seg: 78.1% mAP; Picking point recognition: 88% success rate; Localization error: 2.8511 mm average; Obstruction identification: 100% accuracy.	Segmentation: 88.1% Precision, 86.0% Recall, 87.04 F1-score; Picking point recognition success rate: 88%; Localization average error: 2.8511 mm; Obstruction identification accuracy: 100%; End-effector feeding success rate: 81.3%.

Table A2. Cont.

Authors	Main Goal	Population	Computational Techniques	Gold Standard	Outstanding Accuracy
Sukkuea et al. [46]	Develop a hybrid framework (AGRO-YOLO-V) combining instance segmentation with geometric modeling for accurate single-view orange volume estimation, enabling cost-effective real-time fruit grading in industrial settings.	650 orange samples (Kiew Wan Thai green, Australian citrus, Mandarin) from Southern Thailand; volumes ranging from 50–280 mL; diameters 50–85 mm; 150 held-out test samples.	AGRO-YOLO-V (modified YOLO with Faster-Net blocks and Dynamic Convolution) for instance segmentation; area-to-volume scaling law ($V \propto A^{3/2}$); log-domain variance stabilization; monocular camera with checkerboard calibration.	Bounding-Box Area Calibrated: 12.24% MAPE, 0.89 R ² ; Manual Geometric Baseline (sphere): 13.45% MAPE, 0.82 R ² ; SOLOv2: 8.45% MAPE; Mask R-CNN: 7.12% MAPE.	Mask-Area Calibrated (Proposed): 7.39% MAPE, 0.94 R ² , 9.87 mL MAE, 16.80 mL RMSE; 1.66× error reduction over BBox baseline; 45.1% improvement over manual methods; 14 ms inference latency at 64 FPS; cross-fruit generalization on mangosteen: 8.21% MAPE, 0.92 R ² .
Shi et al. [45]	Develop a lightweight blueberry maturity detection model (BR-DETR-Prune) based on RT-DETR for edge computing environments, addressing dense fruit occlusion, complex backgrounds, and limited computational resources in natural orchards.	6296 blueberry fruit targets (2292 fully ripe, 2174 semi-ripe, 1830 immature) from 2330 images (847 field-captured + 1483 web-mined) of Northern High-bush blueberries from Baishan City, Jilin Province, China; 3 maturity stages based on BBCH scale visual phenotypes.	BR-DETR-Prune (RT-DETR-based) with Faster-Net backbone (PConv), AIFI-RepBN encoder (re-parameterized Batch Normalization), Gather-and-Distribute (GD) feature fusion neck, and Random channel pruning (Lottery Ticket Hypothesis); TensorRT FP16 quantization for edge deployment on NVIDIA Jetson Orin Nano.	Original RT-DETR: 94.4% mAP@0.5, 71.1% mAP@0.5:0.95, 19.88 M params, 57.0 GFLOPs; RTDETR-R50: 94.8% mAP; YOLOv8m: 92.64% mAP, 78.7 GFLOPs; YOLO11m: 93.5% mAP.	BR-DETR-Prune: 97.1% mAP@0.5 (+2.7% over baseline), 74.5% mAP@0.5:0.95, 15.52 M params (−22%), 34.0 GFLOPs (−40%); 69.5 FPS on GPU; 20.5 FPS real-time inference on NVIDIA Jetson Orin Nano Super (FP16).
Zhao et al. [56]	Develop a flexible robot system combining machine vision and Vis/NIR spectroscopy for automated detection and classification of apple internal quality (Soluble Solids Content, SSC), enabling simultaneous grasping, spectral detection, and grade classification.	220 Fuji apples without defects; 180 samples for calibration/cross-validation (harvested from Qixia and Xianyang); 40 samples for independent validation (harvested from Yantai); diameters 75–92 mm; SSC threshold 13.5% for 2-grade classification.	SSD (Single Shot Multibox Detector) with VGG16 backbone for target detection (0.055 s/frame); Vis/NIR spectroscopy (650–1100 nm) with diffuse transmission; PLS regression; NSR (Normalized Spectral Ratio) preprocessing; CARS (Competitive Adaptive Reweighted Sampling) and VCPA wavelength selection; 6-DOF robotic arm with custom end effector for grasping and spectral collection.	Raw spectra PLS: Rcv = 0.836, RMSECV = 0.878%; Otsu threshold method: inaccurate identification under varying light/complex backgrounds; Manual/portable devices: high labor costs, low efficiency; Online equipment: expensive manufacturing costs.	NSR+CARS PLS model: Rc = 0.979, Rcv = 0.969, RMSEC = 0.335%, RMSECV = 0.385%; Independent validation: 90.0% classification accuracy, Rp = 0.952, RMSEP = 0.393%; System completes full process (detection + classification) in 5.200 s per apple; SSD detection speed: 0.055 s/frame.
Ponce et al. [43]	Develop a computer vision-based methodology for automatic counting and individual mass and size estimation (major/minor axes) of olive fruits from images, enabling accurate non-invasive grading for the table olive industry.	3600 olive fruits from 9 varieties (Arbequina, Arbosana, Picual, Ocal, Changlot Real, Verdial de Huévar, Lechín de Sevilla, 967, 1030) from Huelva, Spain; 900 training samples, 2700 external validation samples; 72 regular images + 45 stress images with stochastic fruit placement and touching occurrences.	Image analysis with HSV color space; Otsu's thresholding; morphological opening; distance transform; marker-controlled watershed segmentation for individual fruit separation; ellipse fitting for axis estimation; pixel-weighting functions (logarithmic and root-based) for mass characterization; linear regression modeling (variety-dependent and variety-independent).	Manual measurement variability (SME): Major axis 0.67–2.21%, Minor axis 0.72–1.96%; Manual caliper-based grading: labor-intensive, subjective, time-consuming; Mechanical grading systems: potential fruit damage, limited feature extraction.	Segmentation accuracy: 99.92% (11,606 fruits detected with only 6 errors); Variety-dependent models: Major axis E < 0.80%, Minor axis E < 1.05%, Mass E < 1.16% per variety; Overall: Major axis RMSE 0.4956 mm, E 0.44%; Minor axis RMSE 0.3543 mm, E 0.34%; Mass (Area _{log3}) RMSE 0.1132 g, E 0.49%.
Moallem et al. [66]	Develop a computer vision-based algorithm for grading Golden Delicious apples into quality categories (healthy/defected and first rank/second rank/rejected) using surface features, with robust stem end and calyx detection to avoid false defects.	120 Golden Delicious apple images (360 × 360 pixels) from Blasco et al. database; 40 first rank, 40 s rank, 40 rejected; plus 24 additional images for stem/calyx validation; K-folding with K = 5 (32 training, 8 tests per fold).	Background removal via histogram-based thresholding; MLP neural network (4 inputs: R, G, B, H; 15 hidden neurons) for pixel-level defect segmentation; Stem end detection: morphological opening + Mahalanobis distance classifier; Calyx detection: K-means clustering (K = 2) on Cb component in YCbCr color space; Feature extraction: 8 statistical (mean/std of R, G, B, H), 5 textural (GLCM contrast, correlation, energy, homogeneity, entropy), 3 geometric (defect ratio, perimeter, medial axes length); Classifiers: SVM, MLP, KNN.	Manual/mechanical grading: subjective, time-consuming, inconsistent; 78% accuracy for Golden Delicious (QDC/MLP); 86% accuracy (thresholding on defect size); 73% accuracy (QDC).	Stem end detection: 81.0% (inside apple), 100% (outside apple); Calyx detection: 94.7%; Two-category grading (healthy/defected): SVM 92.5%, MLP 90.0%, KNN 87.5%; Three-category grading (first rank/second rank/rejected): SVM 89.2%, MLP 86.6%, KNN 85.8%.

Table A2. Cont.

Authors	Main Goal	Population	Computational Techniques	Gold Standard	Outstanding Accuracy
Tempelaere et al. [47]	Evaluate and select a robust and efficient deep learning architecture for binary classification (healthy vs. defect) of Braeburn apples using X-ray radiography, with strong focus on generalizability across orchards and storage conditions for industrial sorting applications.	3 datasets of Braeburn apples: Training set (30 healthy + 30 disordered under N2), Test set 1 (40 healthy + 37 disordered under N2 from another orchard), Test set 2 (26 healthy + 26 disordered under CO2); total 60 training + 129 test samples; X-ray images (1632 × 2000 pixels, resized to 224 × 224); 4 radiographs per fruit.	10 ImageNet-pretrained CNNs and Transformer models fine-tuned: AlexNet, GoogLeNet, VGG16, ResNet18, ResNet50, EfficientNet-B0, EfficientNet-B1, ResNeXt50, ViT-B/16, ConvNeXt-T; 10-fold stratified cross-validation; data augmentation (flipping, affine); binary cross-entropy loss; Adam/AdamW optimizers; early stopping.	Manual batchwise inspection (cutting open sample fruit): high food losses (5–25% of crop), destructive, subjective, only batch-level decisions; Batchwise quality control discards entire batch if defects found in sample.	GoogLeNet: 100% (0% IQR) accuracy on Test set 1 (N2 disorders, different orchard); 82% (8% IQR) on Test set 2 (CO2 disorders); ResNet50/ResNeXt50: similar high performance; GoogLeNet with only 49.7 MB size and 1.5 GFLOPs; Model generalizability demonstrated across orchards and disorder types; Enables sample-by-sample quality control reducing food losses.
Hu et al. [64]	Develop a deep learning-based method for identifying bruised apples using 3D surface mesh information from 3D infrared imaging, converting irregular 3D mesh data into 2D feature maps to enable state-of-the-art CNN models for binary classification (bruised vs. unbruised).	302 Granny Smith apples (200 bruised, 102 unbruised); 3D NIR scans producing triangulated meshes with 100,000 vertices and 200,000 faces per apple; 10-fold cross-validation with 272 training, 30 validation samples.	3D local shape descriptors (MaxC, MinC, MeanC, GauC, ShapEl, CurL, ND, DP, Rad); T1 (spherical) and T2 (range) transforms to convert meshes to 2D feature maps; CNN-ETE (AlexNet-based end-to-end), CNN-PT1 (VGG-19 pre-trained + SVM), CNN-PT2 (Inception-v3 pre-trained + SVM); feature fusion and decision fusion strategies; SVM classifier with RBF kernel; AUC of ROC and PR curves.	Handcrafted feature engineering (vmLBP from previous study): 91.67% accuracy; Manual inspection: labor-intensive, subjective, time-consuming; CNN-PT1 (VGG-19) best single-map: 95.67% accuracy; CNN-ETE best single-map: 97.00% accuracy.	CNN-PT2 (Inception-v3) single-map (MaxC): 97.33% accuracy, 0.9857 TPR, 0.9750 PPV, 0.9799 F1; Fusion model (CNN-ETE-f4 with decision fusion on CurL + ShapEl): 97.67% accuracy, 0.9700 TPR, 0.9952 PPV, 0.9820 F1; Best overall: 97.67% accuracy, 6% improvement over handcrafted vmLBP.
Zhang et al. [55]	Develop and evaluate a compact, high-throughput apple infield grading and sorting system using pitch-variable screw conveyors for singulation/rotation/transportation, low-cost imaging for size/color grading, and paddle sorters for separating fresh vs. processing-grade apples at harvest.	Red Delicious and Golden Delicious apples; 240 apples for repeatability (120 per variety); 240 apples for bruise evaluation (80 per throughput); 960 apples for sorting accuracy (480 single mode + 480 continuous mode); system throughputs: 7.5, 9.0, 10.5 fruit per second (2.5, 3.0, 3.5 fruit per second per line).	Machine vision: CCD color camera (15 fps), FoV 460 × 340 mm; two-class linear-discriminant classifier for fruit segmentation; HSV color space with hue thresholds ($t_1 = 15$, $t_2 = 30$) for red/stripe/green classification; size estimation via stem-calyx orientation detection (max equatorial diameter); OpenCV 3.4.5; Qt 5.10 GUI; gear-tooth speed sensor for real-time tracking; paddle sorter with rotary solenoid (40° stroke, 100 ms cycle).	Manual sorting: labor-intensive, subjective, inconsistent, reduces harvest productivity; Previous rotary/cup sorters: prone to malfunctioning at >6 fruit per second, increased bruising; in-house sorting system: 79% sorting accuracy.	Grading repeatability (intra-lane): RD 92–94%, GD 90–91%; Inter-lane repeatability: >81%; Bruise evaluation: 100% graded Extra Fancy, 55–80% no bruising; Sorting accuracy: >99% for all throughputs (7.5–10.5 fruit per second) and both varieties; System meets commercial requirements for infield sorting with robust performance.
Zhang et al. [54]	Investigate the portability and stability of algorithms from hyperspectral imaging (HIS) to multispectral imaging (MIS) for common defect detection on apples, aiming to promote commercialization of fast, non-destructive multispectral inspection systems.	Fuji apples with 8 types of common defects (wind damage, insect damage, bruises, decay, hail, rusting, spot, scar) plus stem, calyx, and sound tissues; diameters 65–90 mm; 109 samples for HIS validation, 116 samples for MIS validation; hyperspectral images (400–1000 nm).	HIS: 400–1000 nm, 1004 × 1000 pixels; MIS: 5 filters (610, 730, 820, 850, 930 nm); PCA and MNF (unsupervised); PLS regression models for decay detection ($R^2 = 0.9486$, RMSEC = 1.31); Band math ($R_{BM} = (R_{925} - R_{730}) / R_{925} \times 255$); Band ratio; Hand-coded VC+/MFC/OpenCV software.	PCA/MNF: unstable, class labels randomly assigned, affected by illumination and physical variability; cannot distinguish stem/calyx from true defects; Poor portability from HIS to MIS; Unsupervised methods require homogeneous conditions; Machine learning methods: time-consuming, low portability.	HIS detection accuracy: 93.6% (102/109 samples) using band math; MIS detection accuracy: 91.4% (106/116 samples) using same band math method; Band math eliminates uneven lightness distribution and curved surface effects; PLS model successfully detects unobvious decay; Proposed methods show strong portability and stability from HIS to MIS, making illumination and physical variability negligible.

Table A2. Cont.

Authors	Main Goal	Population	Computational Techniques	Gold Standard	Outstanding Accuracy
Dutta et al. [59]	Develop a multi-task deep learning framework (FruitQ-GradeX) for simultaneous fruit classification (4 types) and quality assessment (good/bad) using a multi-headed CNN with explainable AI (Grad-CAM) for transparent, real-time agricultural quality control.	9146 images from FruitNet dataset: 4 Indian fruits (Apple: 2290, Banana: 2200, Guava: 2281, Orange: 2375); Binary quality classes (Good: 4630, Bad: 4516); 80% training, 20% validation; Aggressive data augmentation (rotation $\pm 40^\circ$, shifts, brightness 70–130%, zoom $\pm 30\%$, shear 20%).	Multi-headed CNN with shared backbone; 4 convolutional blocks (32, 64, 128, 256 filters) + GAP; Separate classification heads: Fruit (4-class softmax) and Quality (2-class softmax); Cyclic learning rate (triangular policy, step size = 2000, base = 1×10^{-5} , max = 1×10^{-3}); Weighted loss ($L_{total} = 1.5L_{fruit} + 1.0L_{quality}$); Adam optimizer; Batch Normalization; Dropout (50%); L2 regularization ($\lambda = 0.01$); Grad-CAM for XAI; Streamlit deployment.	Single-task CNN models: ~ 94 – 95% accuracy; Conventional machine learning (hand-crafted features): limited performance under complex backgrounds; YOLOv8: 95.8%; Vision Transformer: 0.97 F1-score; 95% dual-task accuracy; 93% multi-output accuracy.	Fruit classification accuracy: 97.71% (validation); Quality detection accuracy: 99.18% (validation); Precision: Fruit 0.98 macro avg, Quality 0.99 macro avg; Recall: Fruit 0.98, Quality 0.99; Guava: 94% precision; Orange: 100% recall; Confusion: 6% guava-apple misclassification due to lighting; Grad-CAM visualizations provide interpretability; Streamlit deployment enables real-time predictions with XAI.
Kevrešan et al. [65]	Perform a comprehensive patent portfolio analysis to identify trends in sensor technologies for measuring fruit properties and their changes, covering sensor-based measurement of individual parameters, multisensor solutions, and integration with artificial intelligence (AI) tools for fresh produce supply chain management.	189 patents, utility models, and patent applications (67 approved patents, 33 utility models, 89 applications) from 20 patenting authorities; First patent from 1998; China (60%), India (13%), Indonesia (6%), Japan/South Korea (3%), the Americas (<5%), Europe (<2%); Most frequent IPC codes: G01N21 (optical, 32%), G01N33 (30%), G01N27 (electric/electrochemical, 20%); Top patent: RS63664B1 (simple family size 28).	Patent portfolio analysis; Latent Dirichlet Allocation (LDA) topic modeling (attempted but unsuccessful due to diversity); Manual patent classification into 3 groups: (I) Sensors for individual parameters (47.6%), (II) Multisensor for complex properties (39.7%), (III) Sensors with AI (12.7%); IPC code analysis; Simple family size analysis; Patenting authority distribution; Trend analysis by year and document type.	Conventional laboratory methods: destructive, time-consuming, labor-intensive, require specialized equipment; Manual inspection: subjective, inconsistent, prone to human error; Batchwise quality control: high food losses (5–25%); Traditional single-sensor approaches: limited to individual parameters, lack holistic assessment.	Sensor technologies enable real-time, rapid, cost-effective nondestructive measurement; Multisensor solutions for holistic quality/ripeness/freshness assessment; AI integration (computer vision, machine learning, deep learning) enables evidence-based, automated decision-making; Patent applications increasing significantly from 2018 onwards (43 patents in 2022); AI-related patents emerging after 2015 with rapid growth; Optical sensors dominate (32%); Substitution of human decision-making in fresh produce supply chains; Supports Industry 4.0 and 5.0 objectives.
Wakhare et al. [49]	Develop a web-based machine learning tool for detecting diseases in pomegranate fruits (bacterial blight, butterfly pomegranate, bitter rot) using image processing with CNN clustering and SVM classification to improve crop yield and quality.	3976 pomegranate fruit images from Kaggle.com: 4 classes (Bacterial Blight, Butterfly Pomegranate, Bitter Rot, Non-infected); 80% training, 20% testing; Images resized to 224×224 pixels; Features: color, morphology, Color Coherence Vector (CCV).	Image preprocessing (grayscale conversion, sharpening, blurring); Feature extraction (color planes, texture, shape via Convolutional Layer/Kernel); CNN for clustering; SVM for classification (binary classification model with optimal hyperplane, regularization parameter C); Intent search technique for user intention from extracted features; Web-based tool for farmer upload and diagnosis.	Computer Vision: 86.6%; Visible-NIR Spectroscopy + Manifold Learning: 88.9%; Deep Learning: 98.06%; Manual inspection: subjective, time-consuming, labor-intensive, inaccurate classification leading to crop losses.	Proposed Model: 98.38% overall accuracy; Per-disease accuracy: Butterfly Pomegranate 98.3%, Bacterial Blight 98.05%, Bitter Rot 97.98%, Non-infected 99.2%; Outperforms computer vision (86.6%) by 11.78%, spectroscopy (88.9%) by 9.48%, and deep learning baseline (98.06%) by 0.32%; Web-based tool provides timely, accurate information for farmers to take preventive measures.
Thinh et al. [48]	Develop a mango classification system using machine vision and artificial intelligence to classify mangoes by color, volume, size, shape, defects, and especially density (maturity/sweetness indicator) for quality grading according to export standards.	1000 mango samples from 3 main commercial Vietnamese mango varieties (Cat Hoa Loc, Cat Chu, and others); Data collected: mass, length, width, height, volume, density, Brix level, defect area; Camera images (RGB) with CCD camera; Loadcell sensor for weight measurement.	Image processing: RGB to grayscale conversion, binary thresholding (threshold 135), contour detection, defect area calculation; Volume estimation: Equation $V_p = 1.1D^2L\pi/6$; Linear regression: Volume = $3.249 \times \text{Length} + 2.956 \times \text{Width} + 10.155 \times \text{Height} - 1000.959$; Neural Network (ANN) with 1 hidden layer (10 nodes), ReLU activation, Adam optimizer, learning rate decay, PCA for dimensionality reduction; OpenCV (C#/OpenCvSharp); PLC integration.	Manual classification: low accuracy, high costs, health effects, labor-intensive, subjective; Previous fuzzy logic system (Malaysia): 80% accuracy; Machine vision maturity prediction (SVM): 4-class classification; Size-based system using image analysis; Manual Brix measurement (destructive).	ANN Brix prediction: 98% accuracy on test set; Volume estimation: average error within acceptable range (R^2 close to 1); Density determination: best quality mango density range 1.00–1.02; Defect detection via contour algorithm; System can classify mangoes by color, volume, size, shape, defects, and density; Enables automated quality grading for export (GAP standards).

Table A2. Cont.

Authors	Main Goal	Population	Computational Techniques	Gold Standard	Outstanding Accuracy
Ni et al. [68]	Monitor the change process of banana freshness over storage time using transfer learning with GoogLeNet, establishing the relationship between freshness and storage dates for non-destructive, automated fruit quality assessment.	103 bananas of two varieties (Good-farmer: 71, Chiquita: 32); 618 images over 11 days of storage at room temperature; Data augmentation (rotation, translation, mirroring) expanded to 4944 images; 70% training, 30% validation; Also tested on 104 strawberries (312 images).	Transfer learning with GoogLeNet (pre-trained on ImageNet); Global average pooling layer instead of fully connected layer; Fine-tuning: learning rate 0.0001, mini-batch size 10, max epochs 10; MATLAB 2018a Neural Network Toolbox; NVIDIA GTX950 GPU; Grad-CAM visualization; 8 data augmentation techniques; 6-class classification (storage days 1, 3, 5, 7, 9, 11).	Human expert classification: subjective, fatigue-prone, inconsistent accuracy declines over time; Traditional machine learning (SVM, ANN) with handcrafted features: limited accuracy for subtle changes; Electrochemical methods: destructive detection; Chlorophyll fluorescence: destructive.	Banana freshness classification: 98.92% accuracy; 3-day interval accuracy: 99.39%; Confusion matrix: most misclassifications only to adjacent days (e.g., day 3 misclassified as day 1 or 5, not as day 9 or 11); Precision: 98.80%, Sensitivity: 98.38%, F1: 98.60% per class; Strawberry freshness: 92.47% accuracy; Model outperforms human experts significantly; ROC curves show near-perfect classification; Transfer learning effective with small datasets.
Aldakhil and Almutairi [57]	Develop a dual-model same-domain transfer learning framework using EfficientNetV2 for simultaneous multi-fruit classification (6 types) and quality grading (good/bad/mixed), enabling integrated automated fruit assessment for agricultural automation.	FruitNet dataset: 14,700+ images of 6 fruits (apples, bananas, oranges, pomegranates, limes, guavas) with 3 quality classes (good, bad, mixed); Augmented to 30,450 samples; 70% training, 20% validation, 10% test; Images 256 × 256 pixels; Captured under various backgrounds and lighting conditions.	EfficientNetV2-S architecture with MBConv and Fused-MBConv blocks; Same-domain transfer learning (classification → grading); AugMix for class imbalance; CutMix and MixUp augmentations; Progressive learning; ReduceLROnPlateau scheduler; AdamW optimizer; Cross-entropy loss; Google Colab (T4 GPU); Streamlit deployment.	Without same-domain transfer learning: validation accuracy starting at 55% (vs 95.1% with transfer learning); Test set accuracy 98.4%; VGG16: 96.75–99.75%; RNN: 98.47%.	Fruit classification model: 99.49% test accuracy; Fruit quality model: 99.42% test accuracy; Transfer learning improves validation accuracy from 55% to 95.1% in first epoch; Classification: Precision 0.99 macro avg, Recall 0.99, F1 0.99; Quality grading: Precision 0.99, Recall 0.99, F1 0.99; Misclassifications primarily between similar fruits (apples-guavas, limes-oranges); Streamlit application enables real-world inference.
Esfandiari Fard et al. [60]	Develop a multi-task Noisy Vision Transformer (NoisyViT) for simultaneous fruit and vegetable freshness detection (binary: fresh/rotten) and type classification (22 classes), creating a comprehensive merged dataset (Freshness44) to improve generalization and accuracy.	Freshness44 dataset: 53,616 images merged from 5 public datasets (Fresh and Stale, Fruits and Vegetables, Fresh and Rotten Fruits, FruitNet, FruitQ); 22 distinct fruit/vegetable types; 44 classes (fresh/rotten per type); Mixed resolutions (144 × 122 to 8000 × 6000); 60% training, 20% validation, 20% test; RandAugment applied.	Multi-task NoisyViT (Vision Transformer base, 12 layers, patch size 16); Positive noise injection (optimal linear transform $Q_{optimal}$) in latent space for entropy reduction; Two parallel heads: binary freshness classification and 22-class type classification; ImageNet-1K pre-trained; Fine-tuning: learning rate 1e-6, AdamW, cosine scheduler, label-smoothing cross-entropy; 30 epochs; Image resolutions 224 × 224 and 384 × 384; NVIDIA RTX 2080 Ti GPU.	Single-task NoisyViT: 99.59% (224 × 224), 99.75% (384 × 384); Standard ViT (no noise): 99.32%; Multi-task CNN: 93.24% freshness, 88.86% type (16 classes); VGG16: 82.2–98.6%; XGBoost: 93.33%; CNN _{BILSTM} : 97.76%; DenseNet201: 99.26%; Traditional ML (SVM, KNN, Decision Trees): 77%.	Multi-task NoisyViT (224 × 224): 99.60% freshness accuracy, 99.86% type accuracy; Multi-task NoisyViT (384 × 384): 99.65% freshness, 99.84% type; Per-dataset: Fresh and Stale 99.85%, Fruits and Vegetables 99.01%, Fresh and Rotten Fruits 98.96%, FruitNet 99.77%, FruitQ 97.98%; Class-wise precision/recall/F1: 99.69%/99.79%/99.74% (type), 99.59%/99.60%/99.60% (freshness); Outperforms all prior methods across all datasets.
Fan et al. [61]	Develop a CNN-based deep learning architecture combined with a low-cost computer vision system for online detection of defective apples on a 4-lane fruit sorting machine at 5 fruits/s, comparing performance against traditional image processing methods.	300 ‘Fuji’ apples (150 normal, 150 defective with insect damage, physical damage, rottenness, scar); 5,400 images acquired (6 images per fruit); 3,300 images selected (1,650 normal, 1,650 defective); Augmented to 79,200 images (rotation 90°/180°/270°, brightness/hue adjustments); 80% training, 20% testing; Independent validation: 200 apples.	Custom CNN architecture: 4 convolutional layers (3 × 3 kernels, 16/64/16/1 filters), 3 pooling layers, GAP layer, softmax; ELU activation; Batch Normalization; Exponential decay learning rate (initial 0.001); Stochastic mini-batch gradient descent with Nesterov momentum; TensorFlow 1.3.0; Images resized to 80 × 80 × 3; Processing time <72 ms per apple (6 images).	Traditional image processing: lightness correction, candidate region segmentation, SVM classifier with 66 features (6 color + 20 textural per RGB component); Training set: 87.1% accuracy, 90.9% recall, 83.3% specificity; On-line NIR structured light: 90.2% accuracy.	CNN model (validation): 96.5% accuracy, 100.0% recall, 92.9% specificity; Independent online validation (200 apples): 92% accuracy with <72 ms processing time per apple (6 images); CNN outperforms traditional SVM by 9.4% in accuracy; Meets commercial packing line speed requirement (5 fruits/s).

Table A2. Cont.

Authors	Main Goal	Population	Computational Techniques	Gold Standard	Outstanding Accuracy
Fan et al. [62]	Develop an improved YOLOv8-based detection model for young peach fruit during thinning stage, addressing challenges of dense overlapping growth, leaf occlusion, multi-scale targets, and complex natural environments to enable automated robotic fruit thinning.	2386 images of young peaches collected from Shennong Fudi Mao Peach Picking Garden (Luancheng, China) using Canon EOS 700D; Resolution 5184 × 3456; 1693 high-quality images after filtering; Data enhancement via offline slicing (1920 × 1080) expanded to 6895 images; 8:1:1 train/val/test split.	Improved YOLOv8n with: FasterNet backbone (PConv+PWCov for lightweight feature extraction), BiFPN (weighted bidirectional feature pyramid with cross-scale fusion), Wise-IoUv3 + MPDIoU loss function (replaces CIoU); PyTorch 2.2.0; Ubuntu 22.04; Dual RTX 4090 GPUs; Input size 640 × 640; HiResCAM feature visualization.	Baseline YOLOv8n (original dataset): F1 = 0.647, mAP@0.5=0.709, mAP@0.5:0.95=0.398; YOLOv8n with image enhancement (clip): F1 = 0.855, mAP@0.5=0.835, mAP@0.5:0.95=0.558; YOLOv10n: F1 = 0.829, mAP@0.5=0.849; YOLOv11n: F1 = 0.845, mAP@0.5=0.865; RT-DETR: F1 = 0.790, mAP@0.5=0.805.	Improved YOLOv8n (FasterNet+BiFPN+Wise-IoUv3+MPDIoU + image enhancement): F1 = 0.973 (↑50.4%), mAP@0.5=0.984 (↑38.8%), mAP@0.5:0.95=0.889 (↑123% over baseline); Model size = 6.3 MB, GFLOPs = 9.8, FPS = 128.5; Outperforms YOLOv10n, YOLOv11n, and RT-DETR; BiFPN ablation: mAP@0.95 ↑60%; Loss function ablation: Wise-IoUv3+MPDIoU achieves highest mAP@0.95=0.889; Heatmap visualization confirms accurate target localization in complex scenes with occlusion and multi-scale targets.
Shi et al. [44]	Develop a knowledge distillation method based on YOLOv8 (Positive Anchor Area Merge Algorithm) for fruit detection tasks, using a larger teacher model (YOLOv8s) to guide a smaller student model (YOLOv8n) to improve detection accuracy while maintaining low resource consumption for embedded deployment.	Self-constructed fruit dataset: 3000+ images per category (apple, banana, orange); Collected from orchards and farmers' markets across different geographical locations and seasons; Multiple angles, high-resolution images; Data augmentation (rotation, scaling, cropping).	YOLOv8-based knowledge distillation with Parallel Network Architecture (student YOLOv8n + teacher YOLOv8s); Positive Anchor Area Merge Algorithm (4 regions: common area, only student, only teacher, negative area); Distillation loss function with 10 components (classification, bounding box, DFL); Binary cross-entropy, CIoU, DFL loss; sigmoid _r activation; SGD optimizer; Batch size 16; 300 epochs; PyTorch 2.0.1; NVIDIA RTX 3060.	Baseline YOLOv8n: mAP@0.5=99.298%, mAP@0.5:0.95=96.959%, Precision = 99.779%, Recall = 98.837%; YOLOv5n: mAP@0.5=99.146%; YOLOv6n: mAP@0.5=99.132%; YOLOv8s: mAP@0.5=99.346%, mAP@0.5:0.95=98.615%.	YOLOv8-lite (ours): mAP@0.5=99.469% (+0.171% over YOLOv8n), mAP@0.5:0.95=96.94%; Precision = 99.969%; Recall = 99.354% (+0.517%); Model size = 24 MB, Params = 3.15 M, FLOPs = 8.7 G, Mean Inf. Speed = 11.7 ms; Faster convergence (peak at 150 vs 200–250 epochs); Raspberry Pi 4B deployment: ~90 ms per frame; Outperforms YOLOv5n, YOLOv6n, YOLOv8n, and comparable to YOLOv8s with 73% fewer parameters.
Hosainpour et al. [63]	Develop a novel machine vision system combining image processing and artificial intelligence for rapid, low-cost, non-destructive quality grading of dried white mulberry (<i>Morus alba</i> L.) into high-quality and low-quality grades.	100 dried white mulberry samples (50 high-quality, 50 low-quality) from local market in Marivan City, Kurdistan, Iran; Manually graded by expert panel based on appearance (color, damaged/broken components); RGB images captured with 13 MP digital camera.	Image preprocessing: RGB to binary masks (BI = R − 0.5 G − 0.5 B); 7 color spaces (RGB, L*a*b*, HSV, NRGB, CrCb, I12I3, gray level) with 19 channels; 10 color features (mean, median, std, CV, kurtosis, skewness, etc.) + 5 GLCM texture features (energy, entropy, correlation, homogeneity, contrast) = 285 total features; Quadratic Sequential Feature Selection identified 23 optimum features; ANN (23-14-2 structure, tangent sigmoid + pure line) and SVM classifiers; MATLAB 2016a.	Manual visual inspection by skilled personnel: time-consuming, expensive, subjective, inconsistent; No prior machine vision system for dried mulberry quality assessment existed in the literature.	ANN (23-14-2): 100% CCR for training, validation, testing, and total data; R = 1 for training/validation/total, R=0.985 for testing; MSE = 0.000005 (training), 0.00066 (validation), 0.01 (testing); SVM: 100% CCR for training, testing, and total; MSE = 0.013 (training), 0.040 (testing); Both classifiers achieved 100% accuracy; ANN superior due to lower MSE; 23 optimum features with distinct mean values between grades; First successful quality grading system for dried mulberry.
Yu et al. [53]	Develop a real-time visual localization system for picking points on strawberry stems for a ridge-planting strawberry harvesting robot, using a rotated YOLO (R-YOLO) detector with rotated bounding boxes to estimate fruit pose and improve harvesting success rate.	2000 strawberry images (1900 training, 100 testing) from ridge-planting cultivation; 573 ripe fruits + 315 unripe fruits in test set; Images captured under varying light, occlusion, overlap, and proximity conditions; Raspberry Pi 3B+ main controller + NVIDIA Jetson TX2 embedded platform; 6-DOF robotic arm; USB camera (640 × 480).	R-YOLO (rotated YOLO-V3) with MobileNet-V1 lightweight backbone (depthwise separable convolution); Rotated bounding boxes with rotation angle α (−90° to 90°); K-means clustering for 9 anchors (3 sizes × 6 angles); Loss function: smooth L1 for α , MSE for w, h , binary cross-entropy for x, y , confidence, class; Transfer learning from COCO; Data augmentation (brightness/contrast ±50%); NVIDIA Jetson TX2 inference (18 FPS).	Original YOLO-V3: harvesting success rate 72.74%; Horizontal bounding box localization: average error >4 mm; Machine vision methods: susceptible to occlusion/overlap, poor robustness; Faster R-CNN: F1 = 0.838; Mask R-CNN: accuracy 95.78% but slower (0.125 s/image); LeNet: accuracy 89.29% at 0.27 s/image.	R-YOLO detection: Precision 94.43%, Recall 93.46%; Ripe fruit Precision 97.17%, Recall 95.81%; Processing speed: 18 FPS on TX2 (0.056 s/image), 3.6× faster than YOLO-V3; Picking point localization average error: ±2 mm (50% improvement over horizontal bbox); Field harvesting success rate: 84.35% (+11.61% over YOLO); Rotated bounding box accurately predicts fruit axis direction; End-effector with laser beam sensors enables contactless harvesting without depth measurement.

Table A2. Cont.

Authors	Main Goal	Population	Computational Techniques	Gold Standard	Outstanding Accuracy
Srinivasagan et al. [69]	Develop a low-cost, portable TinyML-based shelf life estimator (SSLED) for fresh date fruits using Vis-NIR spectral sensors and regression models, enabling real-time, non-destructive prediction of freshness index for supply chain stakeholders.	Khalas cv. fresh dates (Khalal stage) harvested from Date Palm Research Center, King Faisal University, Saudi Arabia; Samples stored under 4 packaging conditions (Control/unsealed, Vacuum Sealed Bags, MAP1 with 20% CO ₂ + N balance, MAP2 with 10% O ₂ + 20% CO ₂ + N balance) at 5 °C and 24 °C; Physicochemical attributes: MC, TSS, sugar, tannin, pH, firmness; Measured every 7 days throughout storage.	AS7265x 18-channel Vis-SWNIR spectral sensor (410–940 nm, 20 nm FWHM); Arduino Nano 33 BLE Sense microcontroller; Edge Impulse Cloud platform for training and deployment; Neural Network regression (2 hidden layers: 20 and 10 neurons, ReLU activation, Adam optimizer, MSE loss); TinyML with TensorFlow Lite for Microcontrollers; Model parameters: batch size 32, learning rate 0.005, 100 epochs, dropout 0.2; Inference time < 100 ms, RAM 1.8 KB, Flash 10.9 KB.	Conventional laboratory methods: destructive, time-consuming, labor-intensive; Manual visual inspection: subjective, inconsistent; Cloud-based ML models: require internet, high latency, bandwidth costs; Shelf life without packaging: 10–12 days at uncontrolled environment; Traditional quality testing requires sample destruction and trained personnel.	Vacuum packaging at 5 °C: shelf life extended to 53 days; MAP1 (20% CO ₂) at 5 °C: 44 days; MAP2 at 5 °C: 23–44 days; TinyML regression model accuracy: 93–100% (confidence threshold 1.5); MAP2 (24 °C): 97.13% MAPE, MSE 0.15; MAP1 (24 °C): 76.4% MAPE; VSB (5 °C): 89.39–98.3% MAPE; Unsealed (5 °C): 75.2–95.2% MAPE; Model correctly predicts shelf life in days (0 = fresh Khalal to fully ripened Tamr); Deployment on Arduino Nano 33 BLE enables real-time handheld device; Battery life 1 week (Cr2032, 225 mAh).
Mputu et al. [67]	Develop a hybrid model combining pre-trained CNNs for feature extraction with traditional machine learning classifiers (SVM, RF, KNN) for tomato quality grading (binary: healthy/reject; multiclass: ripe/unripe/reject) using images captured under uncontrolled lighting conditions.	600 tomatoes collected from local market; 2400 images (4 sides per tomato: top, bottom, front, rear) at 640 × 480 resolution; NVIDIA Jetson TX1 single-board computer; Dataset split: 70% training, 10% validation, 20% testing; Binary dataset: 1600 healthy, 800 reject; Multiclass dataset: ripe, unripe, reject; Public dataset: ripe, unripe, old, damaged (256 × 256).	Image preprocessing: background cancellation (Otsu thresholding on R/G channels, OR fusion, morphological operations); Data augmentation (rotation, reflection, translation, scaling); Transfer learning with 4 pre-trained CNNs (Inceptionv3, ResNet50, MobileNetv2, AlexNet) for feature extraction; Fine-tuning (freezing layers, batch normalization, dropout); 3 traditional classifiers: SVM, RF, KNN; Feature extraction from deepest layers; PCA for dimensionality reduction (KNN); MATLAB R2022b; NVIDIA GeForce RTX 2080 Titan.	Transfer learning only (Softmax): binary 95.52%, multiclass 94.95%; Lightweight CNNs (ShuffleNet, MobileNetv2, EfficientNetB0): 92–95%; ResNet50-SVM: 95.42% binary, 94.25% multiclass; Resnet18: 93.75% binary, 92.50% multiclass; VGG19: 94.17% binary, 93.13% multiclass.	CNN-SVM (Inceptionv3 feature extractor): Binary classification 97.50% accuracy, 98.12% recall, 98.12% precision, 96.25% specificity, 98.12% F1; Multiclass classification 96.67% accuracy, 96.67% recall, 96.74% precision, 98.28% specificity, 96.66% F1; Public dataset (4 classes): 97.54% accuracy; Training time: 4.45–5.73 s (1680 images); Testing time: 2.43 ms/image; Outperforms SOTA by 1.33–4.05%; CNN-RF: 96.46% binary, 96.25% multiclass; CNN-KNN: 95.63% binary; Inceptionv3 excels for datasets with minimal inter-class color differences and 299 × 299 input resolution.

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